Characterization and Modelling of Thunderstorm Wind Fields: Physical Components, Nonstationary Descriptions and Evidence Boundaries

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摘要

Near-surface wind fields produced by thunderstorm downbursts exhibit spatiotemporal inhomogeneity, rapid directional evolution and finite duration; their physical processes cannot be fully described by a single "downburst model." Decomposing such wind fields into independently analysable physical components and reviewing the configuration boundaries of observation, simulation, experiment and modelling routes for each component is a prerequisite for forming a verifiable synthesis. Based on verifiable source-local evidence, this study divides thunderstorm wind fields into fifteen topics: component vocabulary and comparison rules, storm forcing and outflow generation, radial footprint and vortex geometry, translation and path and duration, vertical profiles and low-level extrema, terrain and roughness and urban geometry and boundary-layer interaction, time-varying mean and direction and deterministic-stochastic decomposition, time-varying variance and turbulence intensity and gust factor and integral scale, spectra and evolutionary spectra and time-frequency structure, coherence and spatial correlation and eigenmodes, multivariate and non-Gaussian and data-driven representations, synthesis comparison of information loss in characterization, evidence route comparison and transfer ceilings, the interface from wind speed to pressure and force and response, and conflicts and unreported fields and research gaps. This study examines in parallel the assumptions of stationarity, piecewise stationarity, quasi-stationarity, uniform modulation and evolutionary spectra, distinguishing information retained and lost by time-domain, frequency-domain and time-frequency-domain formulations, and comparing the configuration boundaries of field observations, meteorological and cloud-microphysics models, computational fluid dynamics (CFD), scaled experiments, analytical and semi-empirical models, stochastic simulation, surrogate and reduced-order models and hybrid routes. Duration-related quantities are further distinguished into incommensurable semantic classes including catalogue record-window classes, lifecycle-stage durations, experimental release and scale mapping, inverse-model time parameters, profile-feature persistence, fixed-station decay intervals, experimental sampling durations, record lengths and averaging windows and segment lengths, analysis record durations, simulation loading durations, and facility windows and transition times. A single explicit non-separable condition operator describes the joint modulation of the baseline profile by terrain, roughness, urban geometry and the atmospheric boundary layer (ABL). The synthesis results show that the dominant uncertainties are not concentrated in any single model form but are distributed across definition choices, normalization denominators, time windows, spatial resolution, similarity conditions and cross-route interfaces. Research gaps identified from this synthesis are confined to verifiable matrix blanks and interface breakpoints and are not elevated to design values, code conclusions or independently tested

unified models. This study proposes that effective comparison among sources should be predicated on common physical components, common configuration identity and common resolution intervals, separately recording observational facts, model assumptions, processing methods and author interpretations, thereby preserving source-local numerical values and their locator information while restricting sources that lack configuration declarations, lack independent testing or can only identify research scope to their corresponding statement levels. This framework can support subsequent data collection, model benchmarking and interface tracking but does not replace independent scientific review and human judgement. For unreported fields, this study maintains blanks and states their implications, without infilling from adjacent sources, common-sense inference or model output.
Keywords: thunderstorm wind; downburst; nonstationary wind field; evolutionary spectrum; time-frequency analysis; spatial correlation; evidence boundary

Contents

1 Component vocabulary, configuration identity and comparison rules 5
1.1 Component decomposition hierarchy and naming of descriptive units 5
1.2 Configuration identity, normalization denominators and window comparability rules 6
1.3 Method-family classification axes and comparison protocol 7
1.3.1 Definition of the eight method-family routes 7
1.3.2 Comparison axes, alignment conditions and not-comparable marking 9

2 Storm-system forcing and outflow generation routes 10
2.1 Cloud-microphysics and precipitation-drag resolving forcing 10
2.2 Cooling-source and parameterized forcing 11
2.3 Impinging-jet and wall-jet surrogate forcing 12
2.3.1 Inlet conditions, similarity parameters and scaling rules 12
2.3.2 Apparatus constraints, start-up unsteadiness and background flow 12
2.4 Comparison of forcing routes and transfer ceiling of generation mechanisms 13

3 Radial footprint, vortex structure and geometric representation 14
3.1 Vortex-ring generation, propagation and secondary vortex structures 14
3.2 Footprint definition, threshold dependence and circumferential nonuniformity 15
3.3 Comparison of geometric-quantity extraction representations 16

4 Translation, path and duration 17
4.1 Translation asymmetry and the superposition assumption 17
4.1.1 Applicability boundary of vector superposition 17
4.1.2 Moving reference frames, density currents and flow-field distortion formulations 18
4.2 Path, centre identification and coordinate transformation 19
4.3 Duration semantic classes, definition conditions and source-local evidence 19

5 Vertical profiles and low-level extrema 26
5.1 Analytical and semi-empirical vertical profile families 26

5.1.1 Shape-function families and parameter constraints . . . 28
5.1.2 Near-wall corrections and nonlinear boundary-layer growth . . . 28
5.2 Profile-maximum height, normalized coordinates and near-ground measurement gaps 30
5.3 Modulation of profiles by translation and lifecycle stage . . . 31

6 Terrain, roughness, urban geometry and boundary-layer interaction 31
6.1 Terrain relief, speed-up ratio and reference-field definition . . . 32
6.2 Roughness and the non-equilibrium wall jet . . . 34
6.3 Urban geometry and coupling with an existing boundary layer . . . 36

7 Time-varying mean, direction and deterministic-stochastic decomposition 37
7.1 Weighting-function families and time-varying mean extraction . . . 37
7.2 Direction evolution and vector-scalar decomposition . . . 38
7.3 Cut-off frequency, adaptive degrees of freedom and decomposition uncertainty . . . 38

8 Time-varying variance, turbulence intensity, gust factor and integral scale 40
8.1 Time-varying variance and reduced turbulence intensity . . . 40
8.2 Gust duration and definition dependence of the gust factor . . . 42
8.3 Directional anisotropy and resolution dependence of integral scales . . . 42

9 Spectra, evolutionary spectra and time-frequency structure 42
9.1 Quasi-stationary segment spectra and reduced spectra . . . 42
9.2 Evolutionary spectra and the uniformly modulated formulation . . . 43
9.3 Wavelets and time-frequency localization . . . 44
9.3.1 Time localization and frequency-resolution trade-off . . . 44
9.3.2 Spurious spectral peaks and coupling with preprocessing . . . 45

10 Coherence, spatial correlation and eigenmodes 45
10.1 Coherence-function forms and fitted parameters . . . 45
10.2 Multi-scale correlation and eigenmode decomposition . . . 46
10.3 Spatial support scales and array-layout constraints . . . 46

11 Multivariate, non-Gaussian and data-driven representations 48
11.1 Multivariate conditional simulation and phase consistency . . . 48
11.2 Non-Gaussian residuals and amplitude modulation . . . 49
11.3 Data-driven, surrogate and reduced-order representations . . . 49

12 Synthesis comparison of information loss in characterization 49
12.1 Information-retention matrix across representation domains . . . 49
12.2 Resolution, filtering and resolvable ceilings . . . 50
12.3 Comparability loss caused by normalization and scaling . . . 52

13 Comparison of evidence routes and transfer ceilings 52

13.1 Field-observation route . . . 52
13.2 Meteorological-numerical and CFD routes . . . 53
13.3 Laboratory physical-simulation route . . . 53
13.4 Analytical, stochastic and surrogate routes . . . 54
13.5 Cross-route transfer ceilings and cross-checking boundaries . . . 54

14 Interface boundary of wind-field descriptors 56
14.1 Wind-field observables required at the interface . . . 56
14.2 Evidence boundary of phase, transient and quasi-steady assumptions . . . 58
14.3 Interface unavailability boundary . . . 58

15 Conflicts, unreported fields and research gaps 59
15.1 Explicit conflict items . . . 59
15.2 Unreported-field matrix . . . 60
15.3 Concluding remarks on verifiable research gaps and boundary statements . . . 61

Appendix A Operational definitions of comparison fields and unavailability boundaries 66

1 Component vocabulary, configuration identity and comparison rules

1.1 Component decomposition hierarchy and naming of descriptive units

Quantitative characterization of thunderstorm downburst near-surface wind fields relies on an operable component vocabulary system that discretizes continuous physical processes into independently analysable descriptive units. Radial velocity profiles, vertical distribution functions, time-varying mean envelopes, residual turbulence spectral densities, spatial coherence functions and gust factors constitute the core components recurring in the current literature. However, different studies adopt significantly different coordinate origins, sampling frequencies, time-averaging windows, spatial aggregation extents and normalization denominators when defining these components, creating systematic difficulties for direct cross-study numerical comparison. Solari et al. (2020) systematically reviewed the principal physical quantities used in downburst wind-field characterization together with their measurement and simulation methods, noting the current lack of a cross-scale universal descriptor framework. The literature covered by that review spans field observations, wind-tunnel experiments, computational fluid dynamics (CFD) simulations and analytical models across multiple evidence routes, yet within each route researchers adopt inconsistent definitions of the "equivalent stationary segment," inconsistent peak-velocity extraction rules and inconsistent computation windows for turbulence statistics.

The choice of coordinate origin is the most fundamental source of ambiguity in component comparison. A cylindrical coordinate system centred on the ground impact point is widely used in axisymmetric models, but when a downburst possesses significant translational velocity, the impact point itself changes with time, causing the physical meaning of "radial distance" to drift. The choice of time-averaging window directly affects the separation of the time-varying mean and the residual. Spatial aggregation extent involves the distinction among point measurements, sampling volumes and area averages; the spatial support definition must be preserved during conversion. The choice of normalization denominator lacks a unified convention across studies, so that dimensionless profile shapes may exhibit systematic bias arising from denominator differences.

These disagreements in comparison rules are not secondary issues of technical detail but key constraints limiting the current capacity for knowledge synthesis. Any effort to establish a cross-configuration downburst wind-field model must first specify under which comparison rules the model possesses validity. Taking vertical profiles as an example, when one study normalizes by the jet exit velocity and another normalizes by the instantaneous peak at a reference height, the resulting dimensionless profiles, even if similar in shape, differ fundamentally in physical meaning and applicability boundaries. In spatial aggregation, the difference between single-point anemometer records and area-averaged radar-retrieved wind speeds is not merely numerical but reflects the degree of turbulence information retained under different spatial filtering scales. In the time domain, some studies use the entire downburst event duration as the window for statistical analysis while others extract only the tens of seconds around the peak, and this window-choice difference directly affects the numerical magnitudes of statistics such as turbulence intensity and gust factor.

The completeness of the component vocabulary system itself also requires scrutiny. The components widely used in the current literature are mainly inherited from the analytical framework for stationary boundary-layer winds. This vocabulary possesses full statistical support when describing stationary ergodic processes, but in nonstationary, inhomogeneous and finite-duration downburst

events, the definitional conditions of these components are not strictly satisfied. Whether new descriptive components specifically designed for nonstationary impinging-jet-class flow fields need to be introduced, or whether existing components remain sufficient after appropriate generalized definitions, is a fundamental question that has not yet been systematically discussed. This recognition constitutes the starting point for discussions in the subsequent sections.

Component-level comparison further requires separating "whether a descriptive quantity exists" from "whether the descriptive quantity can be compared." Even when two studies both report peak wind speed, turbulence intensity or spectral density, as long as coordinate origins, reference velocities, averaging windows and spatial support scales differ, their numerical values do not belong to the same comparison unit. A more robust synthesis approach is to first record the physical component, then record the representation domain, configuration identity, normalization rule, estimation method and evidence route, and finally judge which fields can be aligned. Unreported does not mean that a variable does not exist; it means only that the current source cannot support the corresponding cross-study inference. The comparison framework should also preserve the negative boundaries of sources: when a source has not measured wind direction, scalar velocity results cannot support direction-evolution inferences; when only mean profiles are given, instantaneous extrema cannot be inferred; when only simulation results are reported, grid-resolved structures cannot be called confirmed by natural events. Recording these boundaries together with positive findings prevents the review from inadvertently adding information that the sources did not provide. Sources covering three or more declared components provide structural references for establishing the component hierarchy(Hawbecker et al. 2018; Hangan et al. 2019; Zhang et al. 2019).

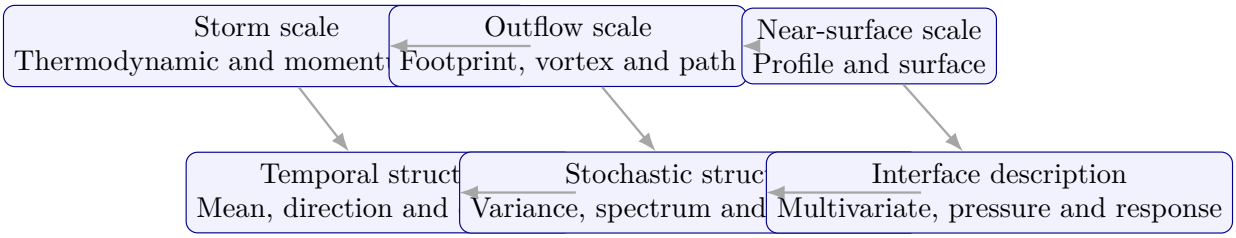


Figure 1: Multi-scale component classification of thunderstorm wind fields. Arrows indicate descriptors that need to be transferred and do not indicate that an independently tested closed causal chain has been established among the components.

Figure 1 presents in conceptual-synthesis form the relationship framework of the component decomposition hierarchy and naming of descriptive units, identifying the decomposition paths from the physical field to each descriptive quantity and the positions of configuration-identity annotations. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

1.2 Configuration identity, normalization denominators and window comparability rules

Configuration identity is the first constraint for judging whether two results enter the same comparison unit. Configuration identity comprises at least a facility or model or event identifier, a scale ratio or full-scale declaration, a jet or storm identity, terrain and roughness conditions, a coordinate origin and reference position, and a normalization denominator. When any one of these fields is missing or inconsistent, the numerical difference between two records may originate from physical mechanisms or from definitional choices and cannot be separated. The difference in

normalization denominators is especially prominent in comparisons of vertical profiles, turbulence intensities and spectral densities. Normalization by jet exit velocity and normalization by ground peak velocity correspond to different dimensionless scales; even when curve collapse is satisfactory, physical equivalence cannot be inferred. Window comparability rules are equally important: the averaging window determines the boundary between mean and residual, thereby altering variance, spectral shape and extrema. When a source does not report window length and type, the subsequent statistics can only be retained as source-local results. Laboratory studies on normalization and scaling conditions provide specific configuration references for such comparisons(McConville et al. 2009; Romanic et al. 2020; Canepa et al. 2022b; Mason et al. 2005), and other sources along the scaled laboratory experiment route also involve configuration identity and window constraints(Lin and Savory 2006). Field-observation route studies involve configuration identity and window constraints(Zhang et al. 2024b; Wang et al. 2026). At the method-genealogy level, related sources are used only to delineate research scope(Žužul et al. 2023; Li et al. 2012) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Wood et al. 2001; Mason et al. 2010a) and not to support specific numerical values, results or method-comparison judgements.

1.3 Method-family classification axes and comparison protocol

1.3.1 Definition of the eight method-family routes

Systematically dividing the methods involved in downburst wind-field research into eight routes is the organizational basis for establishing cross-source comparisons. The first route is field measurement, comprising meteorological towers, anemometer networks, Doppler radars, lidars and rawinsonde records; its advantage lies in preserving the temporal evolution and environmental context of natural events, but it is constrained by station density, instrument response and event rarity. The second route is analytical and semi-empirical models, comprising radial shape functions, vertical shape functions and wall-jet closures; its advantage lies in parameter controllability and computational efficiency, but axisymmetric and steady-state assumptions limit the applicable range. The third route is stochastic simulation, comprising evolutionary spectral representations, modulating functions, autoregressive (AR) and conditional simulation methods; it can generate time histories satisfying target statistical characteristics but depends on prior distribution assumptions and parameter calibration data. The fourth route is meteorological and cooling-source simulation, comprising cloud-resolving models and prescribed cooling-source forcing; it can self-consistently resolve storm forcing and outflow generation but near-surface resolution is constrained by parameterization and grid. The fifth route is Reynolds-averaged Navier-Stokes (RANS), unsteady Reynolds-averaged Navier-Stokes (URANS), large-eddy simulation (LES) and other CFD methods; it can provide detailed spatial flow fields but computational cost, turbulence closure and inlet conditions limit the generalizability of results. The sixth route is wind-tunnel and laboratory physical simulation, comprising impinging jets, wall jets, slot jets, multi-blade devices and actively controlled facilities; it can repeatably change geometric and time-history parameters but similarity conditions and facility constraints limit the transfer range of scaled conclusions. The seventh route is surrogate, reduced-order and data-driven representations, comprising proper orthogonal decomposition (POD), matrix factorization and data-driven generation methods; it can compress information and improve efficiency but the training domain and hold-out scheme constrain extrapolation ability. The eighth route is the wind-speed to pressure and force and response interface route, connecting wind-field

descriptions with structural analysis, but the input observables and assumptions required at the interface are usually not fully matched by the outputs of the preceding seven routes.

Table 1: Comparison of input, output and transfer ceilings of the main method families for thunderstorm wind fields

Method family	Primary input identity	Primary output	Comparison and transfer ceiling
Field observation	Natural events and instrument sampling	Time histories, profiles, spatial samples	Limited by site, sample, data gaps and resolution
Analytical and semi-empirical	Selected parameters and assumptions	Functions, parameters and component relations	Limited by calibration domain, simplifying assumptions and identifiability
Stochastic simulation	Statistics and dependence models	Sample time histories and statistical ensembles	Limited by marginal, phase, non-Gaussian and stationarity assumptions
Meteorological and CFD	Initial-boundary values, grids and closures	Flow fields, spectra and vortex structures	Limited by model, grid, closure and computational domain
Physical simulation	Apparatus inputs and similarity rules	Measurable velocities, pressures and flow fields	Limited by facility, scale, Reynolds number and background flow
Surrogate and reduced-order	Training samples or modal bases	Reconstruction and rapid predictions	Limited by training distribution, truncation and error definition
Velocity interface	Velocity, direction, phase and spatial structure	Input description for pressure or force	Not usable for structural judgement when the input chain is incomplete

Table 1 systematically lists the defining characteristics of the eight method-family routes, including the physical components or processes each route represents, required inputs, delivered outputs, evidence-route identifiers and principal constraints and unavailability boundaries. This table is a synthesis comparison table; each cell uses source-local statements and short qualitative descriptions. Fields marked `not reported` indicate that the current sources have not provided the corresponding information; fields marked `not comparable` indicate that direct side-by-side comparison is not possible due to inconsistencies in configuration, normalization or resolution. This table cannot support route-difference ranking or cross-route testing conclusions.

The multi-dimensional perspective of method-family classification obtains method-genealogy boundary information from the physical-simulation review(Li et al. 2024), impinging-jet experiments(McConville et al. 2009), cooling-source numerical simulations, meteorological observation

networks and review-level syntheses(Solari et al. 2020). At the method-genealogy level, additional related sources are used only to delineate research scope(Vermeire et al. 2011a; Iwashita and Kobayashi 2019) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2012; Wood et al. 2001) and not to support specific numerical values, results or method-comparison judgements. Additional related sources are used only to delineate research scope(Proctor 1988; Mason et al. 2010a; Mitchell and Hovermale 1977; Oreskovic et al. 2018b) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, related sources are used only to delineate research scope(Orf et al. 2012; Oreskovic 2016; Oreskovic et al. 2018a; Zhang et al. 2013) and not to support specific numerical values, results or method-comparison judgements.

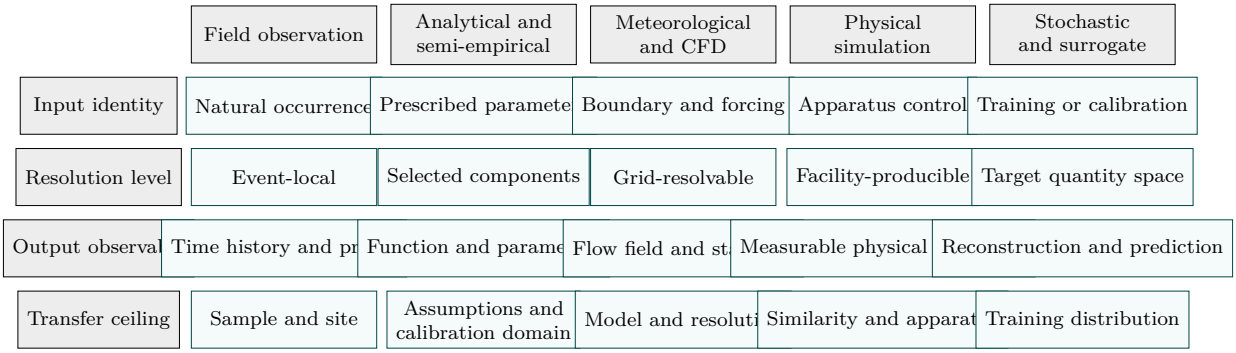


Figure 2: Comparison dimensions of method families. Each column reflects a different evidence identity and information ceiling; no ranking of method superiority exists in the figure.

Figure 2 presents in classification-matrix form the coverage of the eight method-family routes, identifying the position of each route in terms of physical component, representation domain and evidence level. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

1.3.2 Comparison axes, alignment conditions and not-comparable marking

The method comparison in each major science section proceeds along the following mandatory axes: the physical component or process represented, the stationarity assumption, the representation domain (time domain, power spectral density, evolutionary spectra or time-frequency domain), required input data and parameters, output observables delivered downstream, the evidence route, spatial and temporal resolution and near-surface applicability, uncertainty and evidence ceiling, and the unavailability boundary. The minimum condition for two records to enter the same comparison unit is that the component, configuration identity and resolution interval have all been declared. When any one of the three is missing, the corresponding cell is marked `not comparable` with an explanation of the reason. `Not reported` and `not comparable` are strictly distinguished as two different states: the former indicates that the source did not provide the information; the latter indicates that the information exists but cannot be placed side by side because of configuration or definition differences. This distinction is used consistently in the synthesis tables and comparison paragraphs of all subsequent sections. The source(McConville et al. 2009) provides a concrete case of comparison-axis application in the alignment discussion of experimental configurations. At the method-genealogy level, related sources are used only to delineate research scope(Žužul et al. 2023; Li et al. 2012; Wood et al. 2001; Proctor 1988) and not to support specific numerical values, results

or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Mason et al. 2010a,b; Gunter and Schroeder 2015) and not to support specific numerical values, results or method-comparison judgements.

2 Storm-system forcing and outflow generation routes

2.1 Cloud-microphysics and precipitation-drag resolving forcing

The origin of downburst near-surface wind fields lies in the thermodynamic and microphysical processes inside thunderstorm clouds. The intensity, spatial distribution and temporal evolution of the cooling source affect the initial momentum of the downdraft, while the interaction between the downdraft and the environmental atmospheric boundary layer (ABL) controls the spatial structure and temporal characteristics of the near-surface outflow. Hawbecker et al. (2018) used LES to investigate downburst development under different atmospheric stability conditions, reporting a significant modulating effect of atmospheric stability on downburst intensity and spatial extent. Under stable stratification the downdraft is suppressed and the radial expansion range of the near-surface outflow decreases; under unstable stratification, convective mixing enhances the penetrating ability of the downdraft, leading to stronger ground-level peak wind speeds. Burlando et al. (2018) systematically analysed the meteorological characteristics of storm systems producing downbursts based on a Mediterranean thunderstorm event catalogue, observing a statistical association between strong downburst events and specific combinations of environmental parameters. This cataloguing work provides climatological constraints for cooling-source parameterization, but its regional representativeness is limited by the Mediterranean-specific land-sea thermal environment. Butler and Kareem (2007) investigated gust-front propagation mechanisms, reporting the lifting motion and density-current characteristics at the leading edge of gust fronts. Fujita (1990) further supplemented the understanding of the downburst life cycle based on observational data from the Northern Illinois Meteorological Research on Downbursts (NIMROD) programme, recording the temporal wind-speed variation characteristics throughout the complete evolution process from initial descent through ground-level expansion to dissipation. At the method-genealogy level, related sources are used only to delineate research scope(Proctor 1988; Vermeire et al. 2011a) and not to support specific numerical values, results or method-comparison judgements. Additional related sources are used only to delineate research scope(Orf et al. 2012; Oreskovic 2016; Oreskovic et al. 2018a) and not to support specific numerical values, results or method-comparison judgements.

Comparison of forcing mechanisms requires distinguishing three levels. The first level involves the processes producing the downdraft, including in-cloud cooling, precipitation drag and environmental stratification. The second level involves the radial outflow and gust front formed after the downdraft strikes the ground. The third level involves the velocity, direction and turbulence recorded at near-surface measurement stations. Different studies may resolve only one of these levels yet use the same "downburst model" label. If this level is not stated during synthesis, the effect of cooling-source parameters on storm generation may be directly converted to near-surface velocity conclusions.

Hawbecker et al. (2018) investigated downbursts under five atmospheric stability conditions (two convective, neutral, two stable) using LES, reporting ensemble-averaged ground-level maximum horizontal wind speeds in the range 35 to 38 m/s (source text p. 11 table 1), with differences of about 9% among the five configurations. The source also reported minimum vertical

velocities (downdraft intensity) under each configuration ranging from negative 25.98 to negative 29.89 m/s(Hawbecker et al. 2018). The ground-level maximum horizontal wind speed under stable stratification was slightly higher than under convective conditions; this counterintuitive result was attributed by the source to stable stratification weakening convective mixing so that the descending column arrives at the ground more concentratedly. This finding indicates that the modulating direction of atmospheric background conditions on downburst intensity is not monotonic; the simple intuition that "unstable environments produce stronger downbursts" cannot serve as a universal rule. However, this conclusion is limited to the idealized LES configuration and specific cooling-source parameters; actual combinations of environmental wind shear, moisture and microphysical processes may produce different sensitivity directions.

The Mediterranean thunderstorm outflow catalogue of Burlando et al. (2018) classified 277 station records by duration into three classes: 10-minute (50.9%), 1-hour (38.3%) and 10-hour (10.8%), with 186 records in the 15 to 20 m/s peak-velocity interval and only 4 records exceeding 30 m/s (source text). All four records exceeding 30 m/s belong to the 10-minute duration class, but records exceeding 25 m/s span multiple duration classes; therefore no systematic monotonic relationship exists between duration and peak velocity(Burlando et al. 2018). Through meteorological background checks, the source found that some 10-hour duration records actually correspond to extratropical cyclones or frontal systems rather than convective thunderstorms, demonstrating that signal duration and morphology alone cannot serve as the sole basis for event-type classification(Burlando et al. 2018).

2.2 Cooling-source and parameterized forcing

Unlike cloud-microphysics resolving forcing, the cooling-source parameterization method replaces explicit resolution of microphysical processes by prescribing spatial and temporal functions of the cooling rate. The core advantage of this method lies in its ability to run on coarse grids, substantially reducing computational cost while retaining a basic description of the near-surface outflow. The spatial shape of the cooling source is usually simplified to an axisymmetric ellipsoid or cylinder, with its intensity decaying in time according to exponential or cosine functions. This parameterization cannot self-consistently produce the feedback mechanisms in convective processes; therefore, the intensity and structure of the outflow are predetermined by the cooling-source parameters rather than emerging from thermodynamic processes. The parameter choice space of the cooling source is large, comprising cooling-rate amplitude, cooling-zone radius, cooling-zone height, cooling duration, and vertical and horizontal shape functions. Different parameter combinations can produce similar ground-level maximum wind speeds but different spatial distributions and temporal evolutions. Consequently, reporting ground-level peak velocity alone is insufficient to constrain the parameters of a cooling-source model; radial distribution, vertical distribution and time history must be provided simultaneously for multi-dimensional comparison.

When establishing comparability among source routes, it is necessary to distinguish between "the storm system being resolved" and "the outflow being imposed." In cloud-microphysics simulations the outflow emerges from thermodynamic and microphysical processes; in cooling-source models the forcing function is specified by the researcher. Both route types may generate near-surface radial winds yet retain different information about the generation mechanism. A particular advantage of the cooling-source route lies in systematically changing a single parameter and observing the output response, thereby establishing parameter-sensitivity relationships. However, conclu-

sions from such parameter sweeps cannot be extrapolated to conditions where the cooling-source shape assumption does not apply (e.g. strong environmental wind shear, multi-cell interaction or asymmetric cooling distributions). Although the cloud-microphysics route in principle includes more physical processes, its near-surface resolution and parameterization-scheme choices equally constrain output accuracy, particularly in turbulence closure within the boundary layer and in surface-flux parameterization. At the method-genealogy level, related sources are used only to delineate research scope(Mitchell and Hovermale 1977; Oreskovic et al. 2018b; Zhang et al. 2013) and not to support specific numerical values, results or method-comparison judgements.

2.3 Impinging-jet and wall-jet surrogate forcing

2.3.1 Inlet conditions, similarity parameters and scaling rules

Laboratory physical simulations reproduce key features of downburst near-surface outflows through impinging-jet or wall-jet apparatus under controlled conditions. Inlet conditions include jet velocity, jet diameter, nozzle-to-ground distance and velocity programme (steady or pulsed). Mejia et al. (2022) tested five different wall-jet and impinging-jet retrofit schemes in the Wall of Wind (WOW) facility, reporting differences among the schemes in producing downburst-like outflows and noting that the final scheme selection integrates multiple indicators including vortex-ring reproduction, nose-shaped profiles, Reynolds number (Re) and operability. The field observations of Hjelmfelt (1988) provide a full-scale reference for inlet condition selection. The core challenge of similarity parameters lies in simultaneously satisfying geometric, dynamic and temporal similarity conditions, which cannot be fully achieved in most facilities. The nozzle-to-ground-height to nozzle-diameter ratio (H/D) is the key dimensionless parameter controlling vortex sequences and profile morphology. The choice of scaling rules directly affects result transfer from model scale to full scale; Romanic et al. (2020) proposed a scaling method specifically designed for downburst-like impinging-jet outflows. The Wind Engineering, Energy and Environment Dome (WindEEE) facility is widely used in the current literature for downburst-like experiments(Junayed et al. 2019; Canepa et al. 2025, 2022b). At the method-genealogy level, related sources are used only to delineate research scope(Iwashita and Kobayashi 2019) and not to support specific numerical values, results or method-comparison judgements.

2.3.2 Apparatus constraints, start-up unsteadiness and background flow

Physical constraints of the apparatus introduce non-physical features into experimental results; these features need to be distinguished from the true physical processes of downbursts. Li et al. (2021) reported a method for actively simulating transient wind fields by controlling a multi-blade wind-tunnel device via deep reinforcement learning, noting the lag and oscillation problems between the device dynamic response and target wind-field time history. Start-up unsteadiness is another important constraint: in pulsed-jet experiments, the initial transient introduced by the valve opening process may interfere with the arrival of the downburst leading edge. Romanic et al. (2019) observed transient behaviour in impinging-jet experiments in crossflow, reporting systematic effects of background flow on jet trajectory and vortex structure. The presence of background flow breaks the axisymmetric assumption; this has a certain physical correspondence with the behaviour of natural downbursts in environmental wind fields, but the profile characteristics of the background flow in the laboratory usually differ significantly from the natural ABL.

2.4 Comparison of forcing routes and transfer ceiling of generation mechanisms

The cloud-microphysics resolving route, the cooling-source parameterization route and the laboratory surrogate forcing route differ fundamentally in the resolving depth of the generation mechanism. The cloud-microphysics route retains the complete causal chain from thermodynamic forcing to near-surface outflow, but computational cost and resolution limit the number of events that can be simulated and near-surface accuracy. The cooling-source route simplifies the forcing to prescribed functions, reducing computational cost but losing the feedback information between forcing and outflow. The laboratory route replaces meteorological processes with mechanical devices, providing repeatability and parameter controllability, but the limited similarity conditions constrain full-scale transfer of results. The transfer ceilings of the three routes can be stated as follows: the conclusions of the cloud-microphysics route cannot exceed the scales resolvable by its grid resolution and parameterization scheme; the conclusions of the cooling-source route are limited by the form and parameter range of its prescribed functions; the conclusions of the laboratory route are limited by facility geometry, Re and similarity conditions.

Table 2: Comparison of storm forcing and surrogate generation routes

Forcing route	Resolved link	Imposed link	Primary boundary
Cloud microphysics and precipitation drag	Part of storm generation and descent process	Ambient initial-boundary values and microphysics choices	Model closure and resolution
Parameterized cooling source	Forced descent and surface outflow	Cooling shape, intensity and duration	Forcing parameters cannot be directly equated to natural processes
Impinging jet	Touchdown and radial wall jet	Nozzle, inlet and pulse	Apparatus and similarity parameters
Wall or slit jet	Near-wall outflow and shear layer	Inlet profile and channel conditions	Geometric substitution and start-up process
Multi-fan and active control	Tracking of target time history or spatial field	Control target and actuator input	Facility dynamics and training environment

Table 2 compares the differences among the three forcing route types in dimensions including physical-component resolution, inlet conditions, forcing-outflow feedback, near-surface resolution, repeatability and transfer ceiling. In this table, `not reported` marks indicate that a dimension has not yet been sufficiently reported for a given route; `not comparable` marks indicate that direct comparison is not possible in that dimension because the underlying physical mechanisms differ fundamentally. This table cannot support route-difference ranking. At the method-genealogy level, related sources are used only to delineate research scope(Oreskovic et al. 2018b; Orf et al. 2012;

Oreskovic 2016; Oreskovic et al. 2018a) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Zhang et al. 2013; Mitchell and Hovermale 1977; Proctor 1988) and not to support specific numerical values, results or method-comparison judgements.

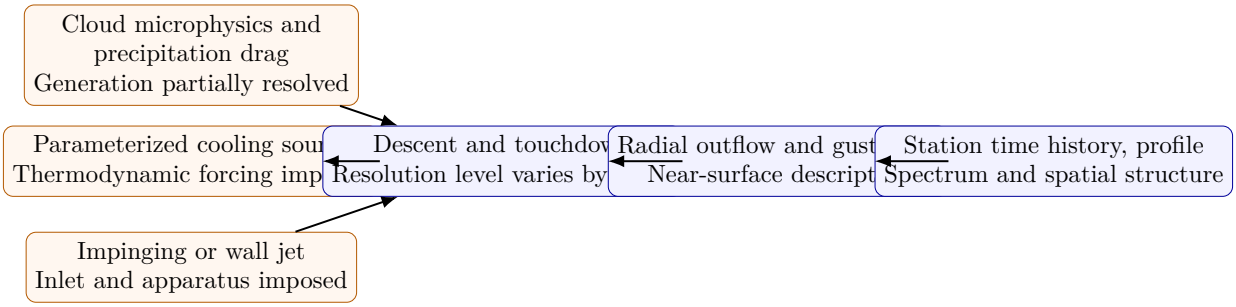


Figure 3: Storm generation and surrogate forcing routes. All three routes can produce near-surface radial outflow descriptions, but they retain different information about the generation mechanism.

Figure 3 presents in conceptual flowchart form the information-transfer paths from inputs to near-surface outflow for the three forcing route types and their respective truncation boundaries. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

Temporal correspondence between storm forcing and near-surface wind also needs to be recorded. Cooling onset, downdraft arrival at the ground, gust-front passage through the measurement station and peak wind-speed occurrence are not simultaneous. If models or observations use different time zeros, phase differences in lifecycle curves may originate from definitions rather than physics.

3 Radial footprint, vortex structure and geometric representation

3.1 Vortex-ring generation, propagation and secondary vortex structures

The radial wall jet formed after a downburst strikes the ground is the direct source of near-surface extreme wind speeds, and the vortex structures accompanying jet development affect the spatial distribution and temporal fluctuation characteristics of wind speed. Vortex-ring generation originates from shear-layer roll-up produced by the collision of the downdraft with the ground; this process forms ring-shaped vortices driven by Kelvin-Helmholtz instability, which subsequently propagate radially outward and undergo stretching, deformation and eventual dissipation. Jesson and Sterling (2018) systematically investigated the parameter sensitivity of downburst vortex-ring dynamics through a parameterized vortex model constructed from superposition of Rankine vortices, using a small number of parameters to describe the position, strength and propagation velocity of primary and secondary vortex rings. The source reported the stretching and deformation characteristics of the primary vortex ring during its radial propagation after ground impact, noting that the primary vortex ring propagates radially outward through the near-wall effect and produces local wind-speed extrema at specific radial distances. The parameterized nature of this model gives it higher computational efficiency than complete CFD solutions, but the Rankine-type vortex assumption produces sharp high-velocity regions whereas the velocity distribution in experimental data is smoother; the source attributed this difference to turbulent mixing during actual vortex interactions. Removing the secondary vortex ring reduced the maximum radial velocity by approximately

11%, from 19.3 m/s to 17.2 m/s (source text pp. 6–9); this quantified difference demonstrates the non-negligible contribution of secondary vortices to near-surface peak velocity(Jesson and Sterling 2018).

Secondary vortex rings are produced by the interaction between the primary vortex ring and the ground boundary layer; their strength is usually weaker than that of the primary vortex ring, but regions through which they pass may experience brief wind-speed enhancement. Junayed et al. (2019) conducted detailed measurements of impinging-jet flow fields for six different H/D and Re configurations in the WindEEE facility, using Cobra probes and two-dimensional particle image velocimetry (PIV) to record the spatiotemporal evolution of vortex sequences, reporting the occurrence patterns of secondary vortices and trailing vortices after the primary vortex ring passes. In configurations with H/D greater than 1, the energy distribution of vortex sequences differs significantly from that in configurations with H/D less than 1. Canepa et al. (2025) further used time-frequency wavelet spectrograms to separate the energy contributions of vortex structures into primary vortex ring, secondary vortex ring and trailing vortex components, reporting that the primary vortex ring energy is concentrated in the 0.5 to 1.5 Hz frequency band whereas the energy produced by secondary vortices shifts to the high-frequency band above 3 Hz. This time-frequency analysis indicates that the characteristics of vortex passage through the measurement point in the time-frequency plane are directly related to vortex propagation velocity and spatial scale.

At the method-genealogy level, related sources are used only to delineate research scope(Žužul et al. 2024) and not to support specific numerical values, results or method-comparison judgements. Zhong et al. (2025) investigated the dynamic impact of vortex structures on a self-supporting tower from the experimental simulation perspective, providing transient force information during vortex passage. Differences in the description of vortex number, strength and propagation characteristics among different sources originate partly from configuration differences (H/D, Re, inlet velocity programme) and partly from detection method differences (probe point measurements, PIV surface measurements, CFD full-field post-processing). When synthesizing these sources, the facility configuration and detection method of each result should be retained, and vortex sequences from specific configurations should not be elevated to universal rules. At the method-genealogy level, related sources are used only to delineate research scope(Childs et al. 2021; Orf et al. 2014; Orwig and Schroeder 2007) and not to support specific numerical values, results or method-comparison judgements.

3.2 Footprint definition, threshold dependence and circumferential nonuniformity

Radial footprint, maximum wind-speed position and vortex structure are related, but they are not interchangeable descriptive quantities. The footprint is a spatial region under a given threshold and observation height; the maximum wind-speed position depends on reference coordinates and extraction time; vortex structure requires velocity-gradient or vortex-identification methods. Axisymmetric images may compress circumferential differences, while instantaneous images may misrepresent brief vortex passage as stable geometry. Geometric synthesis should simultaneously record radial position, circumferential angle, observation height, temporal phase and identification algorithm. Only when these conditions are aligned do the differences among analytical vortex models, numerical flow fields and experimental measurements acquire unambiguous physical meaning.

The northern Mediterranean thunderstorm outflow dataset of Canepa et al. (2024a) provides

99 ten-hour high-frequency wind-speed records that display typical kinematic characteristics of footprint passage through a station: the rapid ramp-up in wind speed corresponds to the passage of the primary vortex ring leading edge, possibly followed by a relatively steady wake segment. However, single-point records can only capture the temporal profile of the footprint at the station location, not the spatial footprint itself. Using Doppler radar, Hjelmfelt (1988) observed downburst morphology in Colorado, showing that the footprint shapes of individual microbursts and microburst lines are largely controlled by the environmental wind field and the spatial distribution of the downdraft; circumferential nonuniformity in natural events is the norm rather than the exception. Laboratory impinging jets, because of the axisymmetry of the inlet conditions, tend to produce relatively uniform circumferential distributions, but facility constraints (such as the hexagonal chamber geometry of WindEEE or the rectangular cross-section of WOW) still introduce circumferential asymmetry. Wang et al. (2026) investigated the spatiotemporal characteristics of downburst-like wind fields through experimental and numerical methods, providing paired observations for the temporal evolution of footprints. At the method-genealogy level, related sources are used only to delineate research scope(Gunter 2015; Vermeire et al. 2011b) and not to support specific numerical values, results or method-comparison judgements.

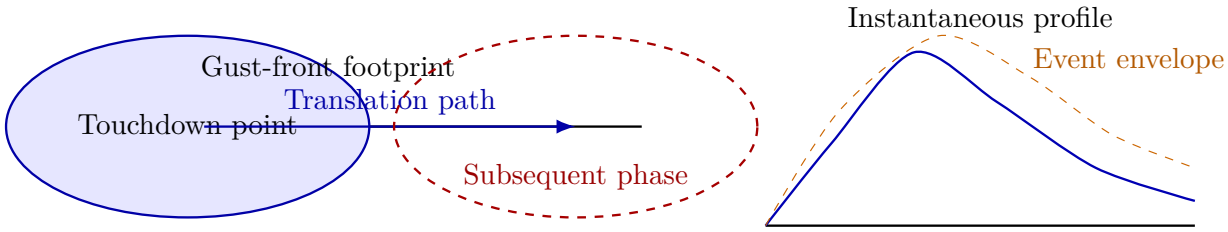


Figure 4: Conceptual relationship among outflow footprint, path and temporal semantics. Instantaneous fields, phase-averaged fields and event envelopes cannot be interchanged without aligning temporal definitions.

Figure 4 presents in conceptual schematic form the variation of radial footprints under different threshold and observation-height definitions, as well as the effect of circumferential nonuniformity on footprint representation. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

The temporal dependence of geometric quantities further requires specifying whether the instantaneous, phase-averaged or event-envelope representation is adopted. Instantaneous footprints can preserve vortex passage and local peaks but are sensitive to sampling time; phase averaging is suitable for repeatable experiments but depends on phase definition and repeatability; event envelopes cover the affected region but lose the fact that peaks at different locations do not occur simultaneously. The three representations answer different questions and should not be merged simply because their graphical appearances are similar. Circumferential nonuniformity makes a single radial profile insufficient to represent the entire footprint. Lin and Savory (2006) provided a geometric-representation perspective different from axisymmetric impinging jets through the two-dimensional slot-jet approach. Density-driven translating downburst experiments(Jariwala 2021) further demonstrated the systematic disruption of footprint symmetry by background flow.

3.3 Comparison of geometric-quantity extraction representations

Systematic differences exist among different routes in how geometric quantities are extracted. Field observations are constrained by station layout and can usually capture only limited spatial profiles of the footprint; PIV and LES can provide two-dimensional or three-dimensional flow-field

snapshots but are constrained by field of view and grid; analytical models give continuous spatial distributions but depend on assumptions such as axisymmetry and stationarity. The parameterized vortex model of Jesson and Sterling (2018) and CFD analysis represent two different strategies for geometric-quantity extraction: the former describes vortex position and strength through a small number of parameters while the latter extracts vortex structures through detailed flow-field post-processing. At the method-genealogy level, additional related sources are used only to delineate research scope(Žužul et al. 2024) and not to support specific numerical values, results or method-comparison judgements. The results of the two approaches may be close under axisymmetric configurations, but under asymmetric conditions the differences reflect information-retention differences of the methods themselves rather than physical differences.

Eissa et al. (2026) used PIV in the WOW facility to obtain the three-dimensional velocity field of wall-jet outflows, distributing the turbulent kinetic energy to each mode through POD decomposition: the first five modes respectively accounted for 42.80%, 22.75%, 5.61%, 3.06% and 2.73% of the turbulent kinetic energy (source text p. 13). This analysis shows that the energy distribution of vortex structures is significantly non-uniform, but the physical interpretation of mode identity requires combination with flow visualization, and the results depend on the chosen decomposition window and dataset. Mason et al. (2005) observed a primary vortex ring, sometimes an intermediate vortex and a larger trailing vortex through pulsed wall-jet experiments, but explicitly did not observe counter-rotating secondary vortex rings in flow visualization; their possible existence was inferred only from pressure results. This observational difference demonstrates that the vortex identification method (flow visualization, pressure field, velocity gradient) directly affects the reported vortex structure.

Another challenge in geometric-quantity extraction lies in the treatment of the temporal dimension. Instantaneous flow fields capture the immediate structure of vortices passing through the measurement plane, but a single instant cannot represent the vortex sequence of the entire event. Phase averaging requires multiple repeat experiments, whose feasibility is limited by the repeatability of laboratory facilities, while natural events are not repeatable. Event-envelope statistics can provide an overall description of the affected range but erase the important fact that peaks at different spatial locations do not occur simultaneously. At the method-genealogy level, related sources are used only to delineate research scope(Childs et al. 2021; Orf et al. 2014; Orwig and Schroeder 2007; Vermeire et al. 2011b) and not to support specific numerical values, results or method-comparison judgements.

4 Translation, path and duration

4.1 Translation asymmetry and the superposition assumption

In many observational and modelling situations downbursts do not strike the ground while stationary but possess significant horizontal translation velocity under the guidance flow of the parent storm. Translation breaks the radial symmetry of the flow field, enhancing radial velocity in the downstream direction and reducing it in the upstream direction.

4.1.1 Applicability boundary of vector superposition

When the ratio of translation velocity to radial outflow velocity is large, the applicability of the linear vector superposition assumption is challenged. Wu et al. (2021) directly compared measured

profiles from moving experiments with the result of adding translation velocity to stationary profiles using a pulsed wall-jet apparatus with both stationary and moving experiments, finding that the curve shape and peak height do not match. In five sets of moving experiments, at translation velocity ratios of Vt/Vjet = 0.09, 0.14 and 0.18, both the normalized peak velocity and peak height exhibited systematic deviations from the prediction of stationary profiles plus translation velocity (source text pp. 10–11)(Wu et al. 2021). The source explicitly stated that the failure of vector superposition may originate from the nonlinear interaction of translation with vortex structure and wall-jet development, rather than a simple velocity-field superposition error. This conclusion is limited to the parameter range of that facility and cannot be directly generalized to other H/D or Re conditions.

Fang et al. (2023) used a movable-nozzle impinging-jet apparatus to test four translation velocity ratio groupings specified by the source, further supporting the existence of nonlinear translation effects from the perspective of wind-field characteristic comparison. The source reported that with increasing translation velocity, the peak wind speed on the downstream side increases while the peak wind speed on the upstream side decreases; this asymmetry cannot be explained by simple translation of the stationary field. Canepa et al. (2023) reproduced translating downbursts in the WindEEE facility through tilted impinging-jet techniques, reporting that the elliptical outflow field produced by the tilted jet cannot be predicted by linear superposition of a stationary axisymmetric field and the environmental boundary layer, particularly in the rear outflow region where the interaction of the primary vortex ring with the reverse environmental flow suppresses vortex expansion. The density-driven translating downburst experiments of Jariwala (2021) provided direct PIV observations from a fluid-mechanics perspective of the influence of environmental ABL shear flow on downdraft column deflection and outflow asymmetry. At the method-genealogy level, related sources are used only to delineate research scope(Fang et al. 2024) and not to support specific numerical values, results or method-comparison judgements.

4.1.2 Moving reference frames, density currents and flow-field distortion formulations

In analytical reconstruction, Xhelaj et al. (2020) and Xhelaj and Burlando (2022) respectively proposed analytical models for downburst wind fields that incorporate translation effects. Xhelaj et al. (2020) adopted a modified radial velocity distribution function, introducing direction-dependent scaling factors to approximate the asymmetry caused by translation, and calibrated the model parameters to two Mediterranean full-scale event records through a two-stage metaheuristic optimization. The key contribution of that source lies in mathematically separating storm translation from the low-level environmental wind(Xhelaj et al. 2020), thereby resolving the difficulty of earlier models in reconstructing the rapid wind-direction change before and after the peak. However, the mean-profile nature of the model means that it does not include turbulence fluctuations; local extrema during transient vortex passage are outside its descriptive range. Xhelaj and Burlando (2022) constructed an inversion framework for downburst geometric and kinematic parameters from the metaheuristic optimization perspective, decomposing the translation effect into a uniform translation component and a flow-field distortion component caused by translation. The source applied differential evolution (DE) and teaching-learning-based optimization (TLBO) algorithms to ten full-scale events, reporting that TLBO converges within approximately 50 to 100 iterations while DE requires close to the maximum limit of 500 iterations to achieve similar stability (source text)(Xhelaj and Burlando 2022). Both methods perform adequately at moderate translation velocities, but both exhibit significant deviations at high translation velocities, indicating that the ability of analytical

models to capture nonlinear translation effects is limited. Fang et al. (2025) further demonstrated from the perspectives of lifecycle stage and azimuthal angle, using a CFD-calibrated semi-empirical model, that the influence of translation on profile morphology is more significant in the dissipation stage and at lateral azimuthal angles than in the mature stage and directly ahead. That source used moving-nozzle LES (Smagorinsky-Lilly subgrid-scale (SGS) closure, translation ratio 0 to 0.3), reporting a trend of increasing frontal peak velocity with decreasing radius as translation increases, while the Holmes model under the same conditions exhibits excessively fast lateral decay(Fang et al. 2025). The density-driven experiments of Jariwala (2021) provided direct observations from a fluid-mechanics perspective of density-current behaviour in a moving reference frame, observing that the environmental ABL shear flow deflects the landing point of the descending column downstream, producing a highly asymmetric outflow field.

4.2 Path, centre identification and coordinate transformation

Path tracking of downbursts depends on the definition and identification method of the centre position. Romanic et al. (2019) analysed jet-trajectory deflection characteristics in impinging-jet experiments in crossflow, reporting the systematic influence of crossflow intensity on the jet landing position and radial expansion direction. The translating downburst experiments of Canepa et al. (2023) provide path and duration data at laboratory scale. Centre-trajectory errors propagate synchronously into radial distance and azimuthal angle; therefore, path information should include at least the path definition, centre identification method, environmental flow conditions and coordinate transformation rules. At the method-genealogy level, related sources are used only to delineate research scope(Abd-Elaal et al. 2013a) and not to support specific numerical values, results or method-comparison judgements.

4.3 Duration semantic classes, definition conditions and source-local evidence

Numerical reports related to "duration" appear widely in the downburst wind-field literature, but these values do not measure the same physical quantity. The time-related quantities reported by different sources may correspond to catalogue record-window classes, lifecycle-stage divisions, laboratory release times, inverse-model parameters, persistence intervals of profile features, fixed-station signal decay intervals, experimental sampling settings, record lengths and averaging windows, analysis record truncations, numerical simulation prescribed durations, and facility operation windows and actuator transition times. The definition conditions, measured objects and physical meanings of these quantities all differ; they cannot be merged, ranked or statistically analysed as a single variable.

Reported duration quantities can be expressed as a functional of recording conditions and definition conditions:

$$\boxed{\text{E1:}\quad D_{\text{reported}} = \mathcal{F}\big(C_{\text{record}},\ C_{\text{definition}}\big)} \tag{1}$$

where D_{reported} is the time-related quantity reported by the source, C_{record} is the recording condition vector (station type, sampling frequency, instrument response, event classification criteria, spatial position) and $C_{\text{definition}}$ is the definition condition vector (class name, threshold choice, averaging window, segmentation rule, normalization scheme). The same physical event produces different D_{reported} under different C_{record} and $C_{\text{definition}}$; therefore this quantity is not an intrinsic

property of the event.

An effective comparison between time-related quantities from two sources requires the following condition to be satisfied:

$$\boxed{\text{E2:}\quad \text{comparable}(D_i,\, D_j) \iff \text{class}(D_i) = \text{class}(D_j) \;\wedge\; \text{aligned}(C_{\text{def},i},\, C_{\text{def},j})} \tag{2}$$

where class is the class label (one of eight semantic classes) and aligned is the definition-condition alignment predicate. Pairs of values not satisfying this condition do not enter the same comparison unit.

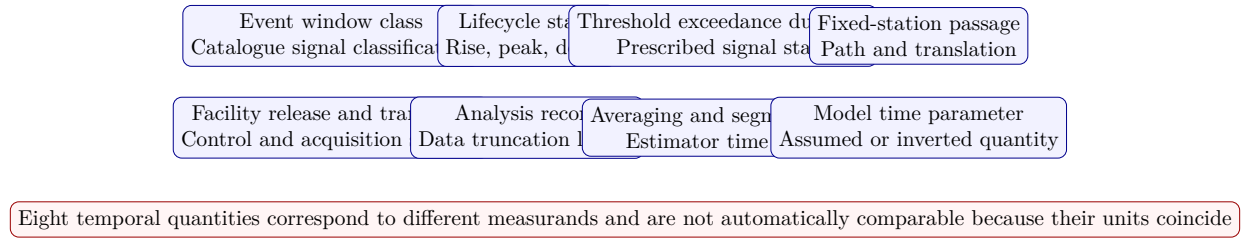


Figure 5: Eight-category classification of duration semantics. Each temporal quantity class corresponds to a different definition, observation or control process; unit coincidence does not constitute a sufficient condition for comparability.

Figure 5 presents in conceptual schematic form the logical relationships among the eight duration semantic classes (catalogue record-window class, lifecycle-stage duration, experimental release and scale mapping, inverse-model time parameter, profile-feature persistence, fixed-station decay interval, experimental sampling duration, record length and segment length), identifying the definition-condition differences and incommensurability among the classes. This figure is a conceptual synthesis and does not indicate that any class possesses priority over others.

The following discusses one by one the eleven time-related quantities reported in the existing sources, grouped by class and accompanied by source-local conditions.

Burlando et al. (2018) classified the detectable duration and signal morphology of 277 Mediterranean station records of thunderstorm outflows into three record-window classes: 10-minute class (141 records, 50.9%), 1-hour class (106 records, 38.3%) and 10-hour class (30 records, 10.8%). This classification is a record-window categorization based on signal characteristics and meteorological scenario review, not a physical storm life span; multi-station records and cases reclassified from extratropical cyclones or frontal systems affected the counts of each class(Burlando et al. 2018). Peak records exceeding 25 m/s span different window classes; therefore no monotonic relationship exists between window class and peak velocity. The evidence ceiling of this source is: a universal duration distribution or a monotonic duration-peak relationship cannot be inferred.

Liu et al. (2023) divided the lifecycle of ten intense thunderstorm outflow events measured at the 356 m Shenzhen meteorological gradient tower into ramp-up phase (mean 194 s), peak phase (mean 1224 s) and dissipation phase (mean 1083 s) using a 60-second moving-average decomposition. The segmentation uses the author-defined phase identification rules; phase lengths depend on event threshold definitions, phase identification rules, tower height, coastal-region sample and averaging window choice(Liu et al. 2023). The means of ten events cannot serve as a universal event template.

Canepa et al. (2023) conducted translating downburst-like experiments in the WindEEE facility through tilted impinging-jet techniques, repeating each configuration ten times, sampling at 2500 Hz, with a release duration of 4.0 s, a normalized time of approximately 10, a source-stated

full-scale mapping to an approximately 10-minute event, and a 0.1-second moving average for signal processing(Canepa et al. 2023). The time quantities in this experiment contain three values with different meanings: the release duration (an experimental setting, not a physical event duration), the normalized time (a dimensionless parameter) and the mapped full-scale event length (dependent on the similarity assumption adopted). The following equation records the source-stated laboratory-to-full-scale time mapping relationship:

$$\boxed{\text{E3:}\quad t_{\text{full-scale}} = t_{\text{lab}} \times \lambda_t \quad \text{(source-stated mapping, not independently computed)}} \tag{3}$$

where λ_t is the source-stated time scaling ratio; this mapping depends on the similarity assumption adopted and cannot be equated with a measured natural event lifespan.

Xhelaj and Burlando (2024) used the TLBO algorithm to invert downburst model parameters, taking a single 40.8 m/s peak event from a Swiss 4 Hz anemometer as the target. After 1024 optimizations, including 100 TLBO iterations for each inversion, the objective function values were distributed between 0.730 and 1.062 (mean 0.893, standard deviation 0.080), and three solution families were identified through cluster analysis. The time parameters of the best solution are $T_{\max}$ = 6.50 min and T_{end} = 29.89 min; the $T_{\max}$ means of the three clusters are respectively 6.50, 13.336 and not reported min, T_{end} means are respectively 29.89, 23.394 and not reported min, translation velocity means are respectively 2.811, 6.527 and 9.307 m/s, and peak radial velocities are 28.639 and 19.006 m/s(Xhelaj and Burlando 2024). These are non-unique inverse-model parameters, not direct duration measurements; geometry, translation and background wind co-vary in the inversion, and parameter-level differences can produce nearly identical single-point ground time histories representing completely different spatial wind fields. The evidence ceiling of this source is: the time parameters of a single cluster cannot be interpreted as a uniquely determined storm duration.

Solari et al. (2020) cited in the review a profile-feature persistence record from a single LiDAR event: the nose-shaped profile remained identifiable for approximately 30 to 40 s, then transitioned to ABL-type growth and levelling profiles(Solari et al. 2020). This profile-feature persistence interval is a profile-morphology persistence time within a single case, not a complete event duration, and as a review citation rather than an original observational report, its evidence identity is limited by the review source chain. The evidence ceiling of this source is: 30 to 40 s cannot be generalized as a universal nose-profile persistence time or event duration.

Fujita (1990) recorded the fixed-station velocity trace (at 16 ft height) of the 2 August 1985 Andrews Air Force Base (AFB) downburst event; the peak reached 149.7 mph (instrument ceiling; the reading was truncated), and from the truncated peak to approximately 2 kt about 2 minutes elapsed, followed by a reverse wind of 84 kt(Fujita 1990). This time interval is the fixed-station signal interval between the truncated peak and the subsequent low-speed state, not a complete event duration; the peak was truncated by the instrument, and path and station position jointly determine the time-history morphology observed at a fixed point. The evidence ceiling of this source is: neither an exact peak nor a universal decay time can be inferred.

Fang et al. (2023) used a movable-nozzle impinging-jet apparatus (nozzle diameter 0.6 m, H/D = 2, platform size 19.5 m × 6 m × 5.4 m, Cobra probes sampling at 256 Hz, Re approximately 4×10^5) for experiments; the single-record length for the stationary condition is 30 s(Fang et al. 2023). This 30 s is an experimental acquisition setting, not an atmospheric event lifespan; facility kinematics and probe position determine the observed passage process. The evidence ceiling of this

source is: 30 s cannot be directly compared with field event stage means.

McCullough et al. (2014) compared fixed-window and adaptive segmentation averaging methods on two full-scale nonstationary datasets (one hurricane record of approximately 19 hours, one rear-flank downdraft (RFD) record of approximately 30 minutes). Fixed window lengths include 3600, 600, 60 s and 300, 120, 60 s in two sets; adaptive methods produce segment lengths including 22.4, 123.7, 91.0, 830.8 s and 12.5, 24.0, 16.8, 40.9 s(McCullough et al. 2014). Record duration, averaging window length and algorithm segment length are three time quantities with different meanings: event type, sampling rate and estimator jointly change the values. The evidence ceiling of this source is: adaptive segment lengths cannot be called storm durations.

Chen and Letchford (2005) conducted POD and wavelet processing based on two full-scale convective wind events, using seven towers with synchronous 1 Hz sampling; the analysis record duration is approximately 1800 s(Chen and Letchford 2005). This record length is an analysis truncation window, not an independently identified event lifespan; tower-array layout and truncation window constrain the value. The evidence ceiling of this source is: a 30-minute physical lifespan cannot be inferred.

Hong (2016) proposed an analytical nonstationary wind model, prescribing several simulation durations: 60, 150, 300, 600 and 3600 s for stationary cases; the nonstationary half-sine strong-wind segment does not exceed 200 s(Hong 2016). These are prescribed simulation loading durations, not observed event distributions; waveform, turbulence assumptions and response model jointly control the computed results. The evidence ceiling of this source is: prescribed durations cannot be used as climatological statistics or design values.

Butler et al. (2010) actively simulated gust-front outflow by superimposing a transient acceleration on an established boundary layer, synchronously acquiring surface pressures at 500 Hz. The facility test window is 120 s and the main-flow transition time is 1 to 2 s(Butler et al. 2010). The facility test window and actuator transition time are facility operating parameters, not natural storm lifespans; scale and model position affect the observed passage process. The evidence ceiling of this source is: facility transition times cannot be elevated to field event durations.

Table 3: Source-local duration quantities and their comparability boundaries

Source and route	Reported time quantity	Definition and configuration	Comparison boundary and locator
Burlando et al. (2018); field catalogue	10-min class 141 records; 1-h class 106 records; 10-h class 30 records	Signal feature and weather-scenario review; record-window classes, not physical storm life	Cannot infer universal duration distribution or monotonic duration-peak relation
Liu et al. (2023); gradient tower	Rise stage 194 s; peak stage 1224 s; decay stage 1083 s	60-s moving average; author-defined stage identification	Lifecycle stages and averaging windows must be separated

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(continued)

Source and route	Reported time quantity	Definition and configuration	Comparison boundary and locator
Canepa et al. (2023); WindEEE experiment	Release 4.0 s; normalized time about 10; source-stated full-scale mapping about 10 min event	2500 Hz; tilted impinging jet; ten repeats per configuration	Facility release time and source-stated scale mapping are not observed lifetime
Xhelaj and Burlando (2024); inverse model	Best solution T_{max} = 6.50 min and T_{end} = 29.89 min; three clusters	TLBO; single 40.8 m/s event at 4 Hz; 1024 optimizations with 100 iterations each; objective 0.730 to 1.062	Non-unique model parameters, not direct measurement or unique solution
Solari et al. (2020); project review	Nose-shaped profile maintained about 30 to 40 s	LiDAR single case cited in review; not original observation	Not event lifetime or population statistic
Fujita (1990); historical trajectory	From censored peak to near 2 kt about 2 min; 84 kt reverse wind	16-ft height; peak censored at instrument limit 149.7 mph	Fixed-station trajectory interval, not complete lifetime
Fang et al. (2023); movable-nozzle experiment	Stationary record 30 s	0.6 m nozzle; $H/D = 2$; Cobra at 256 Hz; Re $\approx 4 \times 10^5$	Acquisition and averaging setting, not atmospheric duration
McCullough et al. (2014); field record method	Hurricane about 19 h; RFD about 30 min; fixed windows 3600, 600, 60 s and 300, 120, 60 s; adaptive segments 22.4, 123.7, 91.0, 830.8 s and 12.5, 24.0, 16.8, 40.9 s	Two full-scale records; fixed and adaptive averaging	Record length, averaging window and segment length are different quantities
Chen and Letchford (2005); tower array record	Analysis record about 1800 s	Seven towers at 1 Hz; POD and wavelet	Analysis truncation window, not independently identified event lifetime
Hong (2016); analytical simulation	Stationary 60, 150, 300, 600 and 3600 s; nonstationary half-sine strong-wind segment not exceeding 200 s	Prescribed waveform; turbulence assumption; response model	Model input, not climatological statistic
Butler et al. (2010); multi-fan experiment	Test window 120 s; main-flow transition 1 to 2 s	Transient acceleration on established ABL; 500 Hz surface pressure	Facility and actuator time, not natural event lifetime

Table 3 uses sources as rows and classes as columns to summarize the time-related quantities reported by the eleven sources discussed above. Each cell records the value, unit or scale, definition and method, measurement or model context, and the reason why direct comparison is not possible. This table does not perform merging, ranking or statistical operations across rows or columns.

Table 4 lists the definition conditions and comparison prerequisites of the eight duration semantic classes: catalogue record-window class, lifecycle-stage duration, experimental release and scale mapping, inverse-model time parameter, profile-feature persistence, fixed-station decay interval, experimental sampling duration, record length and averaging window and segment length. Each class is accompanied by a class label, example sources and source-local statement ceiling.

Catalogue windows and lifecycle segmentation can share the "minute" unit, but the former is primarily a recording and event screening category while the latter is a within-event phase determined by smoothing and thresholds. The 10-minute, 1-hour and 10-hour classes of Burlando et al. (2018) are used for cataloguing; the 194, 1224 and 1083 s values of Liu et al. (2023) are respectively the phase means of ten events. The counting units, denominators and identification rules of the two sets of numbers differ; therefore, catalogue class boundaries cannot be treated as ceilings for phase means, and physical lifespans of records within a catalogue class cannot be inferred from phase means. The 30 to 40 s nose-profile persistence interval cited by Solari et al. (2020) and the approximately 2-minute interval from truncated peak to near-low-speed state of Fujita (1990) belong to yet different classes: the former describes an identifiable profile morphology while the latter describes local signal change after a path passes through a station.

Time quantities in experimental facilities further require distinguishing actuation history, sampling window and scale mapping. The 4.0 s release, 0.1 s moving average and source-stated approximately 10-minute full-scale mapping of Canepa et al. (2023) are not interchangeable; the 30 s record of Fang et al. (2023) and the 120 s test window of Butler et al. (2010) respectively constrain data acquisition while the 1 to 2 s transition time describes facility actuation. If the same apparatus changes travel speed, probe position or background flow while keeping the fixed record length unchanged, the proportion of gust-front leading edge and wake seen by the probe changes. Therefore, facility windows can only be used to describe data coverage and cannot serve as evidence for natural event lifespans or stage durations.

Analysis and model time parameters reside at another level. The time parameters of Xhelaj and Burlando (2024) are jointly inverted with geometry, translation and background wind; multiple parameter families can produce nearly identical single-point time histories; the fixed-window and adaptive segment lengths of McCullough et al. (2014) are estimator choices; the approximately 1800 s of Chen and Letchford (2005) is an analysis truncation; the 60 to 3600 s and less-than-200 s segments of Hong (2016) are prescribed model durations. The three types of values respectively correspond to inversion degrees of freedom, estimator degrees of freedom and simulation settings. Only when class, time zero, window, threshold, spatial position and scaling assumption are all aligned can duration differences be further discussed as event differences; the current frozen sources are insufficient to complete such full-field alignment.

Consequently, statements of "longer" or "shorter" hold only within the same class, common definitions and common spatial support. If two values come from different classes, the correct synthesis result is not a ratio or ranking but a record of the definition breakpoint. Table 3 retains source conditions, Table 4 retains class rules, and Figures 5 and 6 respectively display semantic classes and condition chains; together the four prevent unit consistency from being misread as measurand consistency.

Table 4: Definitions and minimum alignment conditions of duration classes

Time-quantity class	Measurand	Minimum alignment condition	Prohibited merging
Catalogue record-window class	Detectable duration interval into which a station record is classified	Detection and classification rules; counting unit; network and weather screening	Must not be merged with physical lifetime or single-event stage length
Lifecycle-stage duration	Stage length of rise, peak or decay within a single event	Stage threshold; time origin; smoothing and segmentation method; sample definition	Must not be generalized to a universal storm template
Experimental release and scale mapping	Facility actuation time and source-stated full-scale mapping	Facility kinematics; geometric and temporal similarity criteria; sampling settings	Must not be treated as observed natural lifetime
Inverse-model time parameter	Time parameter fitted by a spatiotemporal model	Model form; objective function; parameter constraints; spatial geometry and translation speed	Must not be treated as direct measurement or unique solution
Profile-feature persistence	Time interval over which a specific profile shape remains identifiable	Profile definition; height coverage; number of cases; instrument resolution	Must not be equated to complete event lifetime
Fixed-station passage or decay interval	Local signal interval produced by a moving wind field passing a station	Path; station position; threshold; instrument limit; background wind	Must not be interpreted as entire storm lifetime
Facility sampling and actuation time	Facility record length, release or transition time and averaging window	Sampling rate; actuator timing; station position; number of repeats	Must not be merged with atmospheric event statistics
Analysis, segment and simulation duration	Record truncation, algorithm segmentation or prescribed simulation duration	Record coverage; estimator rules; waveform and simulation assumptions	Must not be converted into climatological or design statistics

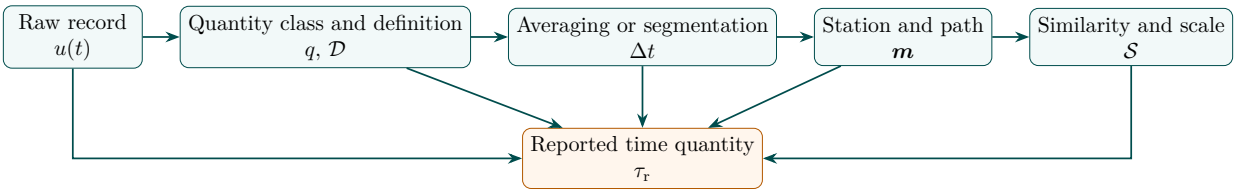


Figure 6: Condition chain of reported time quantities. The raw record, quantity-class definition, averaging or segmentation, station and path, and scale mapping jointly delimit the reported value.

Figure 6 presents in conceptual flowchart form the condition chain from class definitions and recording conditions to reported duration values, identifying the influence paths of definition choices, averaging windows, thresholds, spatial positions and scaling assumptions on the final values. This figure is a conceptual synthesis and does not indicate that any class possesses priority over others.

The eleven sources discussed above cover eight or more temporal semantic classes. The source corpus cannot attribute the observed dispersion of reported time-related quantities to a fixed partition of physical variability and methodological conditions. The relationship between physical duration (if a uniquely definable event lifespan exists) and reported time-related quantities is jointly modulated by class definition, threshold choice, averaging window, spatial position, scale mapping and inversion non-uniqueness. Before the alignment predicate (E2) is satisfied, merging, ranking or performing statistics on time values from different sources does not produce an interpretable synthesis quantity. At the method-genealogy level, related sources are used only to delineate research scope(Fang et al. 2024; Abd-Elaal et al. 2013a) and not to support specific numerical values, results or method-comparison judgements.

5 Vertical profiles and low-level extrema

5.1 Analytical and semi-empirical vertical profile families

One of the most notable features of downburst near-surface wind fields is the fundamental difference between their vertical velocity profiles and the logarithmic-law profile of the ABL. The maximum wind speed occurs at a height of tens to over one hundred metres above the ground, forming a “nose-shaped” profile; this morphology is determined by the wall jet produced when the downdraft strikes the ground. In laboratory impinging jets, the formation of the nose-shaped profile depends on jet velocity, H/D and radial position; in natural events, environmental stratification, surface roughness and storm translation are additional factors. The differences between the two sets of conditions mean that although the “nose-shaped profile” is qualitatively consistent in morphology, the numerical range of peak heights and the characteristics of the sub-peak gradient exhibit significant gaps. Britton and Romanic (2026) developed a downburst vertical profile comparison tool that assembles nine literature profile models and provides a unified graphical interface, reporting differences among the models in peak height, peak velocity ratio and boundary-layer growth rate. The tool covers the Verhoff, Oseguera-Bowles, Vicroy, Holmes-Oliver, Wood and other shape-function families, but its functionality is limited to generating and visualizing the velocity distribution curves of each model at given input parameters and does not include automatic comparison with independent data or statistical testing. Liu et al. (2023) reported, based on measurements from the 356 m

Shenzhen meteorological gradient tower for ten intense thunderstorm outflow events, that ramp-up and peak phase profiles exhibit the nose-shaped characteristic while dissipation phase profiles tend toward the logarithmic-law form of the conventional ABL. These full-scale observations indicate that the peak wind-speed height is not constant during the event lifecycle but changes with time, being typically lower during the ramp-up phase and gradually increasing before dissipation. The measured results support the fitting capability of the Vicroy and Oseguera-Bowles models for the low-altitude ramp-up segment, while the Wood model provides better matching in the high-altitude decay segment. This complementarity among sources demonstrates that a single shape function cannot simultaneously describe the profile morphology precisely over the full height range.

The peak velocity height is not an intrinsic constant of the event but an index quantity that simultaneously depends on configuration, radial position and lifecycle stage:

$$\boxed{\text{E5:}\quad z_{\max} = z_{\max}\big(\text{configuration},\ r/D,\ t/t_{\text{ref}}\big)} \tag{4}$$

where $z_{\max}$ is the profile-maximum height, configuration is the facility or event configuration identity, r/D is the normalized radial position and t/t_{ref} is the normalized lifecycle time. This expression is a definitional declaration, not a fitted relationship.

Table 5: Parameter identity and comparability boundary of vertical profile model families

Profile family	Parameter identity	Peak and near-wall treatment	Comparability boundary
Radial shape function	Peak radius and decay parameter	Pre-peak growth and post-peak decay	Coordinate origin and normalization denominator must be consistent
Vertical shape function	Peak height and shape parameter	Nose-shaped peak and above-peak decay	Near-ground data gaps alter parameter identification
Wall-jet closure	Wall scale and shear-layer parameter	Near-wall gradient controlled by the closure relation	Only conditional analogy with natural events is permissible
Near-wall correction	Roughness or damping parameter	Correction of low-level gradient	Roughness and reference field must be specified
Moving-condition correction	Translation, path or lifecycle parameter	Peak position and shape vary with time	Linear superposition of the stationary field is not treated as fact

Table 5 lists the principal analytical and semi-empirical vertical profile families used in the current literature and their core parameters, including shape-function type, parameter count, peak-height range, normalization denominator and applicability conditions. Some fields are marked `not reported` or `not comparable`. This table cannot support model-difference ranking.

5.1.1 Shape-function families and parameter constraints

The Oseguera-Bowles-Vicroy (OBV) family of models is currently the most widely used analytical profile family; its core idea is to decompose horizontal velocity into the product of a radial shape function and a vertical shape function. Yan et al. (2024a) proposed an improved shape function on this basis, introducing additional parameters for more flexible description of the velocity decay characteristics above the peak height. The vertical shape function of this improved model adopts a power-exponential form in which the power exponent and decay coefficient vary with radial distance to reflect nonlinear boundary-layer growth effects. The experimental configuration is an impinging-jet apparatus with jet diameter 600 mm, nozzle-to-ground distance equal to one diameter and exit velocity 10 m/s on a smooth flat plate (source text p. 3)(Yan et al. 2024a). These configuration conditions limit the transfer range of the calibrated parameters: whether the parameters remain consistent under rough-surface, asymmetric-inlet or higher-Re conditions has not been tested in that source. Abd-Elaal et al. (2013b) further investigated the physical constraints on OBV model parameters, proposing a pair of independent vertical and radial shape functions such that the vertical velocity can be self-consistently derived through the mass continuity equation. That source noted that under certain parameter combinations the model produces non-physical velocity distributions, and demonstrated the improvement of the new function below the peak height through comparisons with Hjelmfelt radar observations, Sengupta-Sarkar experiments and CFD results of Kim-Hangan and McConville et al. However, the radial shape function deviates near the axis from the maximum vertical velocity position predicted by CFD; the authors assigned fitting priority to horizontal velocity because of its dominant role in structural threat. Holmes and Oliver (2000) analysed the effect of nonlinear boundary-layer growth on vertical profiles from a boundary-layer theory perspective, reconstructing moving-downburst event velocity and direction time histories by vector superposition of storm translation, radial impinging jet and environmental background flow. The shape functions of the three sources overlap in functional domain but differ in assumption level: Yan et al. (2024a) directly calibrates parameters from experimental data, Abd-Elaal et al. (2013b) establishes a self-consistent framework from the continuity equation, and Holmes and Oliver (2000) embeds boundary-layer physics. Their normalization denominators and radial applicability intervals also differ; therefore the parameter ranges of the three cannot simply be merged into a single universal parameter table. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2012) and not to support specific numerical values, results or method-comparison judgements.

5.1.2 Near-wall corrections and nonlinear boundary-layer growth

The velocity gradient below the peak height is an independent wind-field quantity in vertical profile characterization; its estimate depends on measurement coverage, near-wall resolution, shape function and calibration configuration. Dang et al. (2024) proposed a piecewise empirical model calibrated against steady-state axisymmetric impinging-jet computations with a Reynolds-stress model, and internally compared the results with radar data, scaled experiments and prior LES and standard k-epsilon turbulence closure (k-epsilon) results. That source uses different functions on either side of the peak height to describe the ramp-up and decay zones. The foregoing comparisons only demonstrate the model performance within the data and configurations selected by the source and do not constitute cross-facility or cross-route independent testing. Holmes and Oliver (2000) discussed nonlinear boundary-layer growth, Abd-Elaal et al. (2013b) constrained shape functions

from continuity conditions, and Yan et al. (2024a) calibrated additional parameters under a specific smooth-surface impinging-jet configuration. The calibration data, turbulence closure, radial interval and normalization denominator of the four routes have not been simultaneously aligned; therefore the existing data can only support source-local profile differences and cannot rank any particular near-wall function as a universal correction. The method-genealogy source(Li et al. 2012) is used only to delineate research scope and not to support numerical or comparison judgements.

Table 6: Configuration identity comparison of near-wall and profile models

Model or route	Configuration identity	Reported profile output	Transfer ceiling
Section 5.1.2 smooth-surface analytical model family	Smooth ground; steady radial outflow; no existing boundary-layer interaction	Parameterized mean velocity profile and source-local peak height	Serves only as baseline configuration identity; not a default input for real surfaces
Canepa et al. (2024b) rough-surface route	Source-specified roughness-element geometry and facility conditions	Profile shape and turbulence quantities under roughness configuration	Must not be transferred to unmatched roughness and scale
Moeini and Romanic (2023) and Moeini (2022) existing boundary-layer routes	Source-specific background boundary layer, symmetry and parameter settings	Source-local peak or profile changes	Different formulations have not been cross-checked under a common configuration
Section 6.1 terrain route	Source-specific terrain and reference field	Fixed-height speed-up ratios or spatial variations	Transfer-context only if profile gradient and peak height are not simultaneously reported
Hadavi and Romanic (2025) and Bélair et al. (2025) urban routes	Idealized urban-canyon numerical configuration and full-scale observation limited by retrieval lower bound respectively	Source-local urban velocity variations or profile above coverage lower bound	Numerical configuration must not supplement observational blind zone

Table 6 lists the configuration-identity correspondence between the near-wall profile models discussed in Section 5.1.2 and the condition factors of Section 6. Each row records the model name or source key, the surface condition assumed (smooth, rough or a specific roughness-element configuration), the ABL-interaction assumption (no environmental wind, linear superposition or nonlinear analysis) and the condition-factor category in Section 6 (direct profile-correction evidence, transfer-context evidence or insufficient evidence). This table is used to track configuration consistency between the baseline profile and the condition corrections.

The foregoing near-wall profile models (the OBV family and its improvements) were all established under smooth-surface or specific calibration conditions, and therefore constitute the baseline reference for the condition-factor discussion in Section 6. The smooth-surface, no-environmental-wind near-wall profile is the starting point of the condition operator; when actual configurations deviate from this baseline, profile morphology cannot be directly superimposed by several independent correction coefficients. Section 6.2 accordingly returns to the configuration identity of Section 5.1.2, distinguishing direct profile evidence for roughness, source-local results for ABL interaction and still-missing joint configurations. The non-separable condition operator E4 and impact matrix T11 preserve this bidirectional connection.

Figure 7 presents in conceptual schematic form the logical relationship between the baseline profile models of Section 5.1.2 and the non-separable condition operator of Section 6, identifying the smooth-surface baseline state, the action paths of each condition factor and the non-independence among factors. This figure is a conceptual synthesis and does not indicate that any condition path

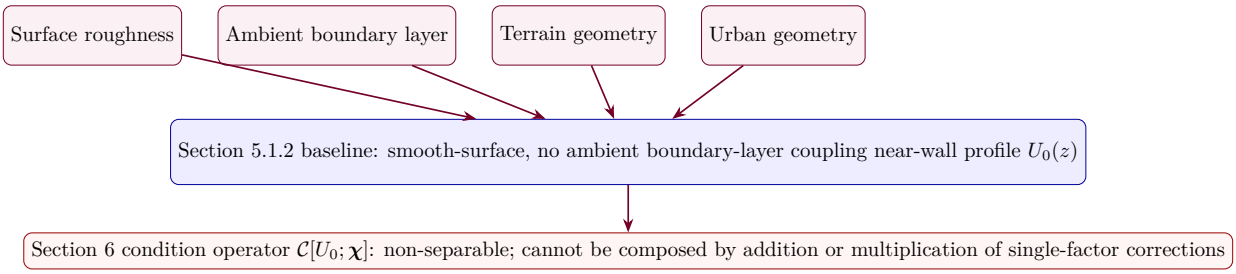


Figure 7: Hierarchical relationship between the Section 5.1.2 baseline profile and Section 6 condition factors. The condition operator remains in non-separable form, indicating that single-factor corrections cannot be directly superposed.

has undergone independent testing.

5.2 Profile-maximum height, normalized coordinates and near-ground measurement gaps

The comparison tool of Britton and Romanic (2026) clearly displays the magnitude of the divergence among different models in peak-height prediction. Differences among models in peak-height prediction can reach several-fold. Yan et al. (2024a) and Dang et al. (2024) respectively corrected the OBV family from different directions, but the two have not yet been integrated within a single unified framework. The height coordinate of vertical profiles also exhibits the distinction among physical height, model-scale height and characteristic-height normalization. Normalization by peak height facilitates shape comparison but conceals the event-to-event differences in peak height itself; normalization by facility or jet scale facilitates experimental similarity but does not automatically correspond to a natural atmospheric scale. Synthesis should simultaneously preserve both the physical coordinate and the dimensionless coordinate.

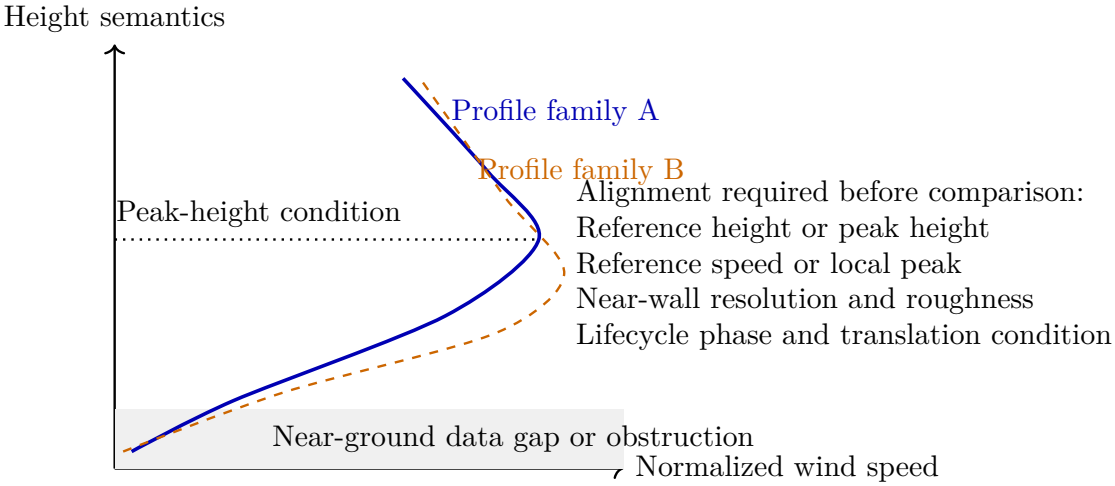


Figure 8: Vertical profile families and near-surface resolution boundary. Curves express qualitative shapes only and do not represent universal profiles or determined parameters.

Figure 8 presents in conceptual schematic form the morphological differences of vertical profile families under different normalization schemes, identifying the definition dependence of peak height and the near-ground measurement gap regions. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

Near-ground measurement gaps are a problem prevalent in field observations and some ex-

periments. When the peak height falls below the instrument installation height or within the roughness-element canopy, the reported "peak" is actually the maximum within the observable region rather than the true physical extreme. Ibrahim et al. (2020) reported in WindEEE facility-constrained LES that the peak height was obscured by the roughness-block canopy and could not be detected (source text p. 7); this finding indicates that the reported value of peak height must be interpreted together with the spatial resolution and coverage of the observation or simulation. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2012; Mason et al. 2009) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Mason et al. 2010a; Canepa et al. 2020; He et al. 2024) and not to support specific numerical values, results or method-comparison judgements.

5.3 Modulation of profiles by translation and lifecycle stage

Fang et al. (2025) specifically investigated the influence of translation on vertical profiles, reporting that on the downstream side of a translating downburst, the peak wind-speed height decreases and the peak velocity increases, while on the upstream side, the peak height increases and the peak velocity decreases. The time-varying model of Pan et al. (2025) parameterizes the profile changes across different lifecycle stages. The definition of lifecycle stages itself depends on the time zero and phase identification rules. At the method-genealogy level, related sources are used only to delineate research scope(Mason et al. 2010a, 2009; Li et al. 2012) and not to support specific numerical values, results or method-comparison judgements.

6 Terrain, roughness, urban geometry and boundary-layer interaction

The near-wall profile models discussed in Section 5.1.2 were all established under smooth-surface or specific calibration conditions; they constitute the baseline reference for the condition factors discussed in this section. The joint modulation of the baseline profile by terrain, roughness, urban geometry and ABL interaction is represented through the following non-separable condition operator:

$$\boxed{\text{E4:}\quad U(z;\,\mathbf{f}) = \mathcal{C}\big[U_0(z);\ f_{\text{terrain}},\ f_{\text{rough}},\ f_{\text{urban}},\ f_{\text{ABL}}\big]} \tag{5}$$

where $U_0(z)$ is the baseline vertical profile under smooth-surface, no-environmental-wind conditions (corresponding to the model family in Section 5.1.2), $\mathbf{f} = (f_{\text{terrain}}, f_{\text{rough}}, f_{\text{urban}}, f_{\text{ABL}})$ is the condition-factor vector and $\mathcal{C}$ is the non-separable condition operator. This operator cannot be decomposed into a product or sum of individual factors because no source in the current frozen corpus has jointly resolved the complete factor vector. Each source resolves only one or two components of the factor vector while keeping the other components at the calibration configuration state. Therefore the corrections are neither additive nor universal.

Table 7 lists the evidence-classification matrix for the condition factors of Section 6. Each row corresponds to one condition factor (terrain relief, surface roughness, urban geometry, ABL interaction); each column records the source key, the reported influence direction and numerical range, the evidence category (direct profile-correction evidence, transfer-context evidence, insufficient ev-

Table 7: Evidence classification of Section 6 condition factors on near-surface profiles

Factor and evidence path	Source-reported profile quantity or context	Evidence class	Evidence still needed
Surface roughness; rough-surface tests and models	Near-surface velocity shape and turbulence quantity changes in source-specific roughness configurations	Direct profile-correction evidence	Joint roughness and background-flow tests under a common reference profile
Existing atmospheric boundary layer; analytical and experimental routes	Peak-height and profile-shape changes under specified background-flow conditions	Direct profile-correction evidence	Cross-checking of different formulations under a common configuration
Terrain geometry; hill and mountain routes	Speed-up ratios and fixed-height spatial variations relative to a reference field	Transfer-context evidence	Matched full profiles and peak-height tracking between terrain and flat reference
Idealized urban canyon; numerical route	Source-local vertical ratios and canyon-geometry dependence	Direct profile-correction evidence limited to idealized configurations	Independent matched observations across the near-surface layer
Full-scale urban observation; LiDAR route	Urban flow evolution above the retrieval lower bound	Insufficient evidence below the coverage level	Observations extending below the profile peak with an open-terrain reference

idence) and limiting conditions. Direct profile-correction evidence requires the source to directly report changes in profile quantities (peak height, peak velocity or gradient); transfer-context evidence means the source reports speed-up ratios or velocity changes but does not directly locate the change to a profile quantity; insufficient evidence means the observational coverage of the source has not reached the profile-peak or near-wall region. This table cannot support universal quantification or independent-testing conclusions for the condition factors.

6.1 Terrain relief, speed-up ratio and reference-field definition

The near-surface wind-field characteristics of downbursts are strongly modulated by surface conditions. Unlike studies of terrain effects in conventional boundary-layer winds, the radial wall-jet structure of downbursts means that the vertical profile and approach direction of the incident flow are spatially inhomogeneous; therefore, terrain acceleration and sheltering effects cannot simply adopt the speed-up ratio formulae from boundary-layer wind conditions. The definition of terrain speed-up ratio requires an explicit reference field:

$$\boxed{\text{E6:} \quad S(z) = \frac{U_{\text{terrain}}(z) - U_{\text{ref}}(z)}{U_{\text{ref}}(z)}} \tag{6}$$

where $U_{\text{terrain}}(z)$ is the velocity with terrain present and $U_{\text{ref}}(z)$ is the velocity over a flat surface under the same incident conditions. $S(z)$ is a transfer-context quantity, not a direct profile correction: the speed-up ratio describes the relative velocity change but does not directly give the change in profile-peak height or shape. Only when the source simultaneously reports both the reference-field definition and changes in profile quantities can the speed-up ratio be converted to profile-correction evidence.

Yuan et al. (2026) investigated downburst flow-field characteristics over mountainous terrain through wind-tunnel experiments, using both wall-jet and impinging-jet apparatus and testing six idealized hill models. The wall-jet apparatus has nozzle height 60 mm, width 1800 mm and exit

velocity 30 m/s(Yuan et al. 2026); the impinging-jet apparatus has diameter 600 mm, nozzle-to-ground distance equal to one diameter and exit velocity 8 m/s (source text pp. 4–5)(Yuan et al. 2026). The source reported a maximum speed-up ratio of approximately 0.33 at the hilltop, maximum deceleration absolute values of 0.62 to 1.0 at the hill foot (depending on slope and azimuth), and terrain influence ranges of approximately 1.5 times the hill height vertically and approximately 4 times the hill height radially (source text p. 10)(Yan et al. 2024b). These quantitative results depend on specific approach flow conditions, geometric scale ratios and hill positions and cannot serve as universal terrain-correction parameters. Yan et al. (2024b) further investigated the effects of different hill slopes and position relationships on downburst-terrain interactions, placing hill models at 40 nozzle heights downstream in the wall jet to ensure that the approach flow is fully developed. The source found that slope and radial position have strong effects on the speed-up ratio while the effect of hill height is relatively weak; two shapes (quadratic and cosine) produce similar results under matched-slope conditions. The source attempted to establish a correspondence between the 40-nozzle-height-downstream position in the wall jet and the 1.5-diameter position in the impinging jet, using the peak velocity height as the matching parameter, but one hill and a single matching method cannot prove universal equivalence(Yan et al. 2023). Yan et al. (2023) extended the research to more complex multi-ridge terrain, observing that when the radial outflow of a downburst traverses multiple ridges, each ridge produces an independent acceleration-deceleration sequence, and the wake effect of upstream ridges affects the approach-flow conditions of downstream ridges.

Table 8: Comparison of correction mechanisms due to terrain, roughness, urban geometry and background boundary layer

Surface mechanism	Reference-field requirement	Primary affected quantity	Not-comparable condition
Terrain relief	Flat or no-terrain co-configured baseline	Path, local acceleration and vertical motion	Reference position or height differs
Surface roughness	Explicit roughness elements and inflow baseline	Near-wall gradient, peak height and turbulence	Roughness and resolution not reported
Urban geometry	Geometry and background boundary-layer baseline	Channel, sheltering, separation and local peaks	Only single-point or single-azimuth evidence
Existing boundary layer	No-background-flow or same-stability baseline	Outflow asymmetry and vertical mixing	Background wind and stability not aligned

Table 8 compares source-local results under different terrain, roughness and urban-geometry configurations, including facility type, terrain parameters, roughness conditions, reported speed-up ratio or velocity change and reference-field definition. Some fields are marked `not reported` or `not comparable`. This table cannot support universalized terrain-effect conclusions.

Figure 9 presents in conceptual schematic form the distinction between the terrain speed-up ratio (E6) and profile correction: the speed-up ratio measures velocity change relative to a reference field, while profile correction requires further determining changes in peak height, peak velocity and gradient. The figure identifies the positioning of the speed-up ratio as transfer-context evidence and the additional information needed to move from transfer context to direct profile correction. This figure is a conceptual synthesis and does not indicate that any transfer path has undergone

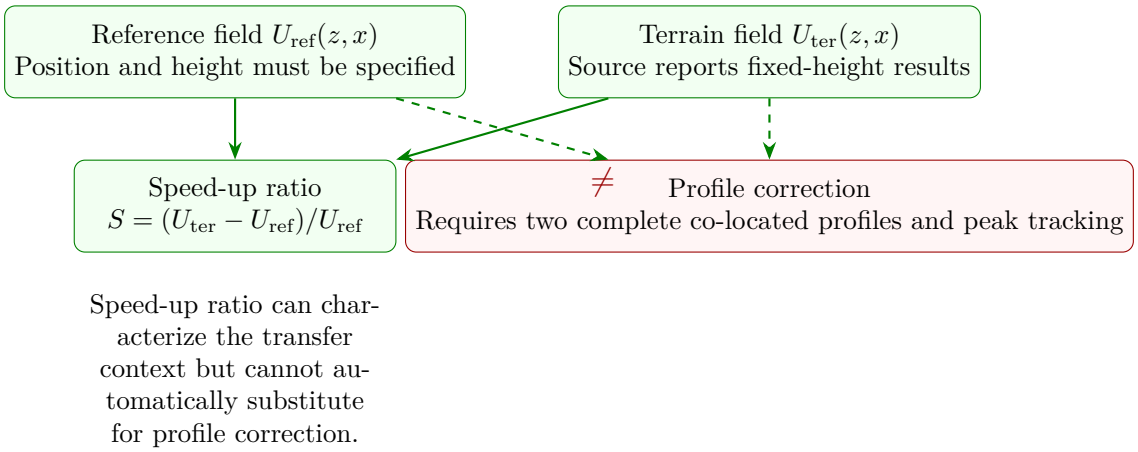


Figure 9: Distinction between terrain speed-up ratio and profile correction. A fixed-height speed-up ratio depends on an explicit reference field and cannot substitute for peak or gradient correction in the absence of paired complete profiles.

independent testing.

Terrain and roughness results also depend on the choice of reference field. Velocity change relative to a flat smooth surface, relative to the same position without terrain, and relative to the background boundary layer have different denominators. If only "speed-up ratio" is reported without preserving the reference, values cannot be compared across sources. At the method-genealogy level, related sources are used only to delineate research scope(Mason et al. 2010b) and not to support specific numerical values, results or method-comparison judgements.

6.2 Roughness and the non-equilibrium wall jet

The near-wall profile models discussed in Section 5.1.2 provide the baseline vertical velocity distribution under smooth-surface conditions. When surface roughness deviates from this baseline, the development characteristics of the wall jet change, but the mechanism of this change differs from the parameterization approach for roughness in conventional boundary-layer winds. In an equilibrium ABL, the roughness effect is parameterized through the displacement height and roughness length of the logarithmic law, and this parameterization assumes that the inner layer has fully adapted to the surface. However, the wall jet of a downburst has not established equilibrium with the ground during its radial propagation; therefore, roughness corrections to the velocity profile cannot be replaced by equilibrium parameterization. The roughness effect and the ABL interaction effect are two separate factor components of the condition operator (E4) and need to be discussed separately.

Canepa et al. (2024b) systematically investigated the development characteristics of wall jets under different roughness conditions. That source used a 3.2 m diameter nozzle in the WindEEE facility, testing two jet velocities of 8.9 and 12.4 m/s corresponding to Re of 1.92 and 2.68 times ten to the sixth power at H/D approximately 1.19, with Cobra probes deployed at ten radial positions and eleven vertical heights sampling at 2500 Hz (source text pp. 4–5)(Canepa et al. 2024b). The source reported a general trend that increasing roughness causes the peak wind-speed height to increase, peak wind speed to decrease and boundary-layer thickness to increase. These trends constitute direct profile-correction evidence (classified as "direct profile correction" in T11), because the source directly reports the change direction and numerical range of profile quantities (peak height and peak

velocity). However, these quantitative relationships cannot be extrapolated from equilibrium ABL roughness-correction formulae, because the wall jet of a downburst is inherently a non-equilibrium flow whose inner-layer thickness has not reached the equilibrium state corresponding to the surface conditions within the finite radial distance.

Moeini and Romanic (2023) investigated the interaction between downburst outflow and ABL winds from the perturbation-analysis perspective, deriving a closed analytical solution. The source reported that under full-scale Joint Airport Weather Studies (JAWS) parameters (radial distance and release height both 2700 m, jet velocity 25 m/s), the nonlinear term contributes negative 3.6 m/s at 273 m height, with its peak relative contribution reaching 44% of the total velocity (source text p. 10); this value indicates that linear superposition is insufficient under these conditions(Moeini and Romanic 2023). This evidence demonstrates that the ABL interaction factor (f_{ABL}) correction to the profile is nonlinear and cannot be independently added to the roughness factor (f_{rough}). Moeini (2022) further established a complete theoretical framework for this interaction from the analytical perspective, reporting that in the given fixed theoretical example, as environmental wind speed increases from zero to 5 m/s, the peak height decreases from 53.6 m to 45.6 m (source text pp. 67–68). Both analytical works jointly demonstrate that the ABL interaction correction to profile morphology is nonlinear and highly coupled, but both are limited to the downstream symmetry plane and the time-independent assumption; transient and three-dimensional effects are not included. At the method-genealogy level, related sources are used only to delineate research scope(Canepa et al. 2022a; Romanic and Hangan 2020) and not to support specific numerical values, results or method-comparison judgements.

Roughness evidence further requires distinguishing "profile quantities are directly reported" from "background flow conditions are changed." Canepa et al. (2024b) directly reports the change directions of peak height, peak velocity and boundary-layer thickness with roughness configuration, and therefore belongs to source-local direct profile evidence in T11; this classification does not transfer the roughness-element geometry, nozzle scale or Re to other facilities. Moeini and Romanic (2023) and Moeini (2022) directly give profile or peak changes within their respective analytical configurations, but both are constrained by steady-state, symmetry-plane and parameter-choice assumptions and cannot be synthesized with roughness experiments as independently addable correction terms. The smooth-surface models of Section 5.1.2 likewise do not provide such a synthesis relationship; their role is only to preserve the baseline configuration identity.

Terrain and urban data have different evidence identities. The terrain speed-up ratios in Section 6.1 use a specific reference field as denominator; if the source does not simultaneously give peak height and gradient changes, it can only serve as transfer-context evidence. Hadavi and Romanic (2025) directly reported source-local velocity changes in an idealized urban-canyon numerical configuration, but the geometric ratio, approach-flow ratio and turbulence closure jointly limit its interpretation. Bélair et al. (2025) provides above-canopy single-event vertical observations in the urban environment; its 49 m installation height and 30 m range gate mean that the near-wall interval is not within the direct coverage, so for that interval only insufficient evidence can be marked. Here "insufficient evidence" means the measurement support domain does not cover the target height and does not imply that the near-ground peak does not exist; it also cannot be filled in from urban-canyon numerical results.

The impact matrix therefore adopts three mutually exclusive labels: direct profile-correction evidence requires the source to report profile quantities such as peak height, peak velocity or gradient; transfer-context evidence only reports relative velocities, speed-up ratios or environmental

conditions; insufficient evidence means the target height or variable has not been directly resolved. The same factor may fall into different categories in different sources; the classification unit is "source and configuration," not mechanism names such as terrain, roughness or urban. No joint experiment or joint computation exists in the current data that simultaneously changes terrain, roughness, urban geometry and ABL while maintaining a common reference field; E4 is therefore only a conditional bookkeeping relationship, not a computable empirical formula.

6.3 Urban geometry and coupling with an existing boundary layer

Characterization of downbursts in urban environments faces additional complexity. Canopy drag produced by building groups and vortex shedding within canyons interact with the wall-jet structure of downbursts, so that near-surface wind-field characteristics in urban environments cannot be simply extrapolated from open-ground experimental results.

Hadavi and Romanic (2025) simulated the interaction between downbursts and urban canyons through RANS CFD. The source tested three ABL velocities (approximately 2.3, 4.5 and 9.0 m/s, corresponding to jet-to-ABL velocity ratios of approximately 20, 10 and 5), reporting that the canyon peak wind speed under weak ABL conditions is approximately three times that under moderate ABL conditions and approximately twice that under strong ABL conditions (source text)(Hadavi and Romanic 2025). In comparisons between low-rise street canyons and open ground, the peak wind speed in low-rise street canyons exceeds that on open ground by more than 70%, and the downstream peak decay is approximately 5% in canyons versus approximately 38% on open ground(Hadavi and Romanic 2025). The peak in high-rise street canyons is approximately 40% higher than in low-rise canyons and more than 130% higher than on open ground(Hadavi and Romanic 2025). These results are all limited to the steady-state impinging jet, specific geometric ratios and the selected RANS turbulence closure (low-Re and standard k-epsilon results are close, realizable k-epsilon overpredicts the peak by approximately 40% at specific locations)(Hadavi and Romanic 2025) and cannot be generalized to arbitrary urban layouts.

Bélair et al. (2025) observed a single severe thunderstorm outflow event on 13 July 2023 using a coherent pulsed Doppler lidar system deployed in the Montreal urban boundary layer. The lidar installation height is 49 m, using an eight-beam velocity azimuth display (VAD) mode with a 30 m range gate, and the analysis is truncated at approximately 1200 m height (source text pp. 4–5)(Bélair et al. 2025). The source recorded the vertical structure and temporal evolution of the above-canopy wind field, observing wind speeds exceeding 40 m/s at approximately 590 m height and the vertical penetration characteristics of the outflow primary vortex(Bélair et al. 2025). However, the spatial resolution and homogeneous-volume assumption of the lidar may not be fully satisfied under urban outflow conditions, and the authors explicitly discussed this limitation. Because of the physical limitations of the installation height (49 m) and range gate (30 m), this lidar system lacks direct observational coverage of the wind field inside the urban canopy below 49 m; therefore it cannot be used to confirm or deny the peak position or near-wall gradient within the canopy. This observation coverage lower limit constitutes the direct basis for classifying the urban-geometry factor as "insufficient evidence" in T11: the profile region below the observation coverage layer lacks direct evidence support, which is a limitation of observational capability rather than proof that the physical quantity does not exist.

Figure 10 presents in conceptual schematic form the relationship between the lidar coverage lower limit (constrained by installation height and range gate) and the identifiability of the profile-

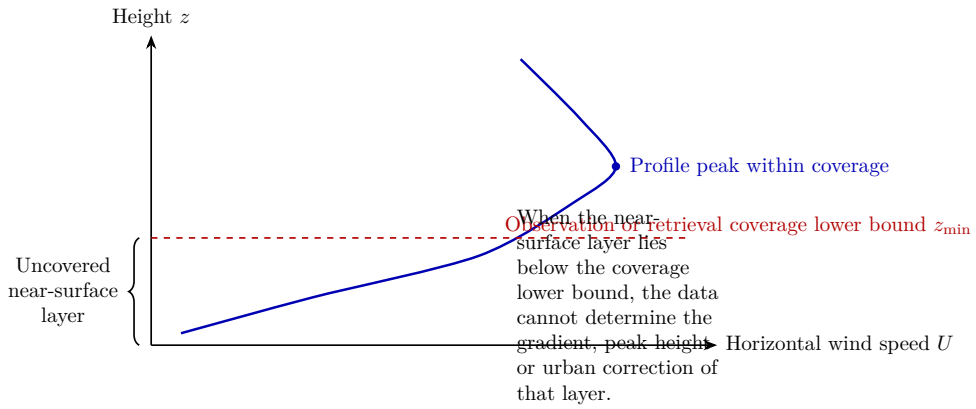


Figure 10: Relationship between observational coverage lower bound and profile-peak identifiability. The near-surface layer below the coverage lower bound cannot be directly supplemented from the profile above.

peak height. When the coverage lower limit is above the possible peak-height region, the observation cannot confirm or deny the existence of the peak; therefore the associated evidence is classified as "insufficient evidence." This figure is a conceptual synthesis and does not indicate that any particular coverage configuration possesses universal representativeness.

Terrain, roughness, urban geometry and the ABL are not mutually independent correction terms. If calibrated separately and then linearly added, the model implicitly assumes no coupling among the mechanisms. No source in the current frozen corpus has jointly resolved the complete factor vector **f** of the condition operator (E4); therefore the corrections are neither additive nor universal. Evidence synthesis is more appropriately conducted by recording the terrain scale, slope, roughness elements, building layout, outflow centre position and environmental flow for each configuration and comparing the mechanisms actually resolved by each source. Sources lacking joint configurations can only support single-mechanism or source-local descriptions. At the method-genealogy level, related sources are used only to delineate research scope(Mason et al. 2010b; Canepa et al. 2022a; Romanic and Hangan 2020) and not to support specific numerical values, results or method-comparison judgements.

7 Time-varying mean, direction and deterministic-stochastic decomposition

7.1 Weighting-function families and time-varying mean extraction

The nonstationary nature of downburst wind speeds renders the traditional signal decomposition framework based on time-averaged values and fluctuation components no longer directly applicable. In stationary boundary-layer winds, the choice of averaging window (typically 10 minutes or 1 hour) does not significantly change the mean estimate because the statistical moments of the process are approximately constant over that scale(Tubino and Solari 2020). However, in downburst events, the mean change from near zero to the peak and back to dissipation may be completed within tens of seconds; the averaging window choice directly determines which changes are assigned to "mean evolution" and which to "turbulence fluctuation," and this decomposition result has no unique physical definition. Tubino and Solari (2020) compared the performance of three methods in downburst signal processing: constant weighting (standard moving average), Gaussian weighting and cardinal-sine weighting. Through analytical derivation of the transfer function of each

weighting function, and using Monte Carlo simulations and 13 Livorno port thunderstorm records as source-internal comparisons, the source attributed the previously observed low-frequency spectral peak to the oscillatory characteristics of the constant-weighting transfer function rather than to an intrinsic event feature (source text p. 14). Gaussian weighting (alpha equal to 12) produces a reduced turbulence spectral density that is smooth and consistent with the band-pass-filtered form of the boundary-layer wind model in the low-frequency range. This finding imposes a methodological constraint on downburst spectral analysis: the low-frequency spectral features reported in studies that use standard moving averages to extract the time-varying mean need to be re-examined. McCullough et al. (2014) proposed the variable averaging interval method for handling the nonstationarity of downburst wind speeds; this method allows the averaging window to vary with time to adapt to the local characteristics of the signal, but its parameter calibration process introduces additional subjective judgement. Su et al. (2015) compared three time-varying mean extraction methods: kernel regression, discrete wavelet transform (DWT) and empirical mode decomposition, finding that different methods applied to the same record produce maximum extreme-value response differences of 12% in the RFD event and 13% in the squall-line event (source text pp. 7–8). That source attributed the differences to the decomposition method; from the perspective of this study, this demonstrates that the wind-field decomposition output from the same record depends on the processing path.

7.2 Direction evolution and vector-scalar decomposition

Zhang et al. (2019) systematically investigated the characteristics of directional decomposition based on 141 isolated downburst event records from a Mediterranean port ultrasonic anemometer network, finding that approximately 40% of the records exhibit a maximum direction deviation of 45 to 90 degrees near the peak, approximately 21% exhibit deviations of 90 to 180 degrees and approximately 10% exhibit deviations exceeding 180 degrees. Under directional decomposition with 30-second moving-average filtering, the longitudinal and lateral turbulence intensities converge to 0.12 and 0.11 respectively, in contrast to the empirical relationship under conventional strong-wind conditions where lateral turbulence intensity is approximately 75% of the longitudinal value(Zhang et al. 2019). This finding suggests that direction effects on nonstationary signal decomposition cannot be ignored.

Directional decomposition can be performed in a fixed geographic coordinate system, a time-rotating along-wind coordinate system or an event principal-axis coordinate system. The coordinate choice alters the component means, variances and cross-correlations. If the coordinate definition is missing from reported decomposition results, subsequent multivariate simulation and coherence comparison lose their common basis.

7.3 Cut-off frequency, adaptive degrees of freedom and decomposition uncertainty

Chen and Letchford (2004) developed a deterministic-stochastic hybrid model to jointly treat the time-varying mean and the random residual. That source assumed that the time-varying mean can be separated into the product of a vertical profile function and a time function, implying that all heights simultaneously reach the maximum mean (source text p. 3). The fluctuation component was assumed to be a Gaussian process with amplitude modulation coefficient equal to 0.25 times the time-varying mean, and the frequency shape was assumed time-invariant (source text pp. 5–6)(Chen

and Letchford 2004). These assumptions provide an operable simplification for the model framework under data-scarce conditions, but each can be replaced by a more refined formulation, reflecting the modular nature of the model rather than the unique determinacy of the parameters. The spatial correlation structure adopts the Davenport coherence function with a vertical decay coefficient of 10; this choice is explicitly a borrowing in the absence of downburst-specific coherence data. Hong (2016) investigated amplitude modulation and frequency modulation phenomena in downburst wind-speed signals from a signal-processing perspective, mapping stationary wind to a nonstationary process through a dimensionless distance-time transformation so that the frequency content retains its dependence on the instantaneous mean. The source reported that the variance of the residual signal exhibits a pronounced clustering effect near the peak wind speed, and demonstrated on an idealized single-degree-of-freedom system that the equivalent gust factor and peak displacement for short-duration nonstationary winds are significantly lower than the corresponding values for stationary winds of the same duration (source text). Chen and Letchford (2007) further explored the frequency cut-off issue in deterministic-stochastic decomposition. The source used a 7-level decomposition with 20th-order Daubechies wavelets (Db20) (equivalent 128-second window) to restrict the mean of two downburst records to below 0.0078 Hz, and reported pulse durations of approximately 170 s for the RFD event and approximately 100 s for the squall-line event (source text p. 3). The source used an empirically fitted AR(2) model to describe the spectral structure of the reduced fluctuation, and reported that the Wood mean profile approximates the estimated profile at given parameters but deviations still exist. The physical basis for cut-off frequency selection is not sufficient in the current literature; the selection ranges of different sources vary from approximately 0.005 Hz to approximately 0.05 Hz, causing the residual energy and mean morphology of the same record to change systematically. Yuan et al. (2024b) reported the evolution characteristics of downburst-like flows produced by an actively controlled multi-blade facility, noting the additional challenges of time-varying mean extraction from the facility's own nonstationary start-up process in the laboratory environment.

Table 9 compares the core parameters, assumptions and influence directions on subsequent statistics of different time-varying mean extraction methods. The methods in this table include moving average, weighted smoothing, adaptive segmentation, wavelet decomposition and empirical mode decomposition. Some fields are marked `not reported`. This table cannot support method-difference ranking.

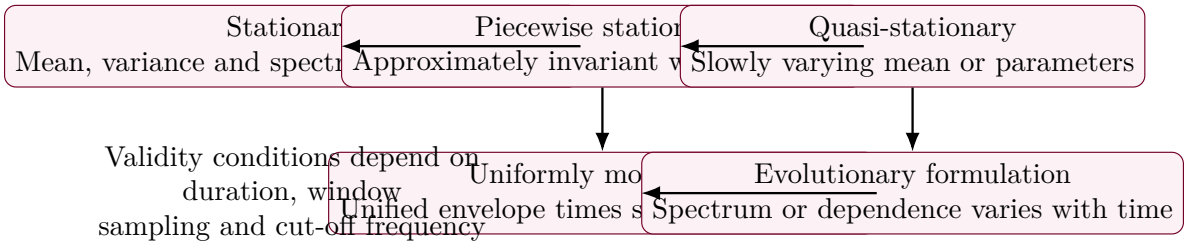


Figure 11: Hierarchy of stationarity and nonstationarity assumptions. The assumption type determines the interpretive scope of statistical quantities and does not imply that any single type holds for all thunderstorm events.

Figure 11 presents in conceptual block-diagram form the propagation of choices at each stage of the deterministic-stochastic decomposition chain on subsequent statistics (variance, turbulence intensity, spectra, extrema). This figure is a conceptual synthesis and does not indicate that any method has undergone independent testing.

Table 9: Comparison of stationarity assumptions and time-varying mean decomposition methods

Decomposition method	Stationarity assumption	Main degrees of freedom	Information effect
Fixed-window averaging	Locally stationary within window	Window length and shape	Alters low-frequency residuals and peak alignment
Cosine or Gaussian weighting	Smoothly time-varying mean	Kernel width and boundary treatment	Transfer function may introduce spurious spectral peaks
Variable averaging interval	Locally varying time scale	Adaptive window	Comparison must preserve the selection criterion
Deterministic-stochastic hybrid	Deterministic envelope plus stochastic residual	Envelope and residual models	Whether peaks are synchronous depends on assumptions
Time-varying autoregressive	Parameters evolve with time	Order and coefficient trajectory	Preserves time-varying dependence but limited by identification stability

The deterministic component and random residual boundary is jointly defined by the analysis method. The window or cut-off frequency determines which variations are interpreted as storm evolution and which as turbulence. The decomposition method is also affected by record boundaries and missing-data treatment. Finite duration causes filter endpoints, empirical decomposition endpoints and wavelet boundaries to produce additional errors.

8 Time-varying variance, turbulence intensity, gust factor and integral scale

8.1 Time-varying variance and reduced turbulence intensity

After the time-varying mean is extracted, the statistical characteristics of the residual signal become the core content of downburst turbulence description. The definition of turbulence intensity in downbursts faces a fundamental difficulty: the conventional definition uses the mean wind speed as the denominator, but when the time-varying mean experiences the progression from near zero to the peak and back during the event lifecycle, the denominator in low-speed segments may cause nonphysically extreme values of turbulence intensity. Chen (2005) observed in vector time-varying autoregressive (TVAR) analysis that when the mean falls near stagnation or reversal points, the inferred time-varying turbulence intensity exhibits extreme maxima; the authors attributed this to the method not capturing the rapid changes in standard deviation (source text pp. 127–128). Huang et al. (2015) reported, based on two downburst records, that turbulence intensity remains at approximately 0.12 for most of the time but deviates from this level at times when the mean is small; the authors attributed these anomalies to a small-denominator effect rather than strong

turbulence itself (source text p. 3). This phenomenon imposes a methodological constraint on how downburst turbulence intensity is reported and compared: the source-local numerical value of turbulence intensity must be interpreted together with the mean-extraction method, window length and denominator definition.

Ibrahim et al. (2020) systematically characterized the turbulence characteristics of downburst wall jets using the WindEEE facility combined with LES. In the WindEEE facility-constrained LES case, the source reported a turbulence intensity of approximately 12% at the peak mean-speed location (source text pp. 7–11); this value is close to the typical values from the full-scale Mediterranean observations cited above, but because of the differences in facility configuration, scale ratio and turbulence generation mechanism, this numerical proximity cannot be interpreted as physical equivalence(Ibrahim et al. 2020). Junayed et al. (2019) measured impinging-jet flow fields for six H/D and Re configurations through a large-scale experimental apparatus, reporting laboratory-scale values of turbulence intensity and integral scale. The source explicitly noted that impinging jets cannot capture the buoyancy-driven characteristics of real downbursts; therefore, its turbulence statistics reflect the vortex-sequence characteristics of the mechanical jet rather than the complete turbulence structure of natural events. Choi (2000) analysed the turbulence characteristics of tropical thunderstorm strong winds using thunderstorm observation data from the Singapore region. That source reported 0.3-second gust factors of 2.6 to 5.8 at 10 m height for thunderstorm winds versus 2.6 to 2.9 for non-thunderstorm winds(Choi 2000); turbulence intensities in all directions are higher than under non-thunderstorm conditions, with differences exceeding a factor of two in some directions (source text table 2). These full-scale observed values reflect the environmental conditions specific to the tropical land-sea boundary; their regional representativeness cannot be extrapolated to other climatic zones. Wang et al. (2026) conducted a combined analysis of downburst-related experimental and simulation results, supplementing source-local turbulence-statistic comparisons from the perspective of paired wind-tunnel and LES results. Zhang et al. (2019) reported, based on directional decomposition analysis of 141 isolated downburst events, ensemble-averaged longitudinal turbulence intensity of 0.12 and lateral turbulence intensity of 0.11 (source text); the near-unity ratio of lateral to longitudinal turbulence intensities contrasts with the empirical relationship in conventional strong winds where the lateral intensity is approximately 75% of the longitudinal value, because the 30-second moving average removes the low-frequency nonstationary trend, so that the analysis variance arises primarily from the isotropic inertial subrange.

Time-varying variance, turbulence intensity, gust factor and integral scale all depend on the mean definition of the preceding section. If the averaging window changes, the residual energy in the numerator and the mean in the denominator both change simultaneously. When sources do not report the window, truncation rules or advection velocity, the corresponding statistics should be retained as not directly comparable. Le and Caracoglia (2019) reported a near-peak-height instantaneous longitudinal turbulence intensity of approximately 0.11 in the multi-blade facility at a maximum deflection angle of 45 degrees (source text p. 10); this value overlaps with the range from full-scale observations, but the facility cross-section dimensions, roughness conditions and inlet turbulence all differ from natural events, so the numerical proximity is a configuration-local result. Yuan et al. (2024a) reported an initial turbulence intensity of approximately 25% under steady-state downburst-like profile conditions in the actively controlled multi-blade facility (source text p. 8); this value is higher than those from the other sources discussed above, reflecting the significant influence of blade-control approach and wind-tunnel flow management on turbulence levels.

8.2 Gust duration and definition dependence of the gust factor

The gust factor divides a short-term peak by a longer average; therefore it simultaneously depends on peak duration and averaging window. In nonstationary events, the denominator varies with time, and a single event-level ratio mixes all lifecycle stages. At the method-genealogy level, related sources are used only to delineate research scope(Kwon and Kareem 2019) and not to support specific numerical values, results or method-comparison judgements. Zhang et al. (2024b) provided source-local gust-factor results for full-scale events based on a Beijing meteorological-tower downburst record. Choi (2000) reported that in tropical thunderstorm records with high-gust events in specific directions, the 0.3-second gust factor can reach 3.7 to 5.8 (source text figure 8), whereas the corresponding values at the same station under non-thunderstorm conditions are only 2.3 to 2.9. Part of this difference may arise from the rapid velocity jump when the gust front passes through the station, causing the ratio between the short-term peak and the longer average to be amplified by nonstationary effects. Wang et al. (2013) reported, in a full-scale data-driven simulation of a rear gust-front event, a 60-second gust-factor mean of approximately 1.35, simulation value 1.33; maximum gust factor measured 1.99, simulated 1.98 (source text table 1 p. 7); this consistency is limited to that specific event and data-driven framework and does not represent testing of a universal gust-factor model. A more transparent approach is to separately preserve the peak extraction, mean envelope and local ratio, accompanied by the instrument response and event segmentation method. At the method-genealogy level, related sources are used only to delineate research scope(Bowen 1996) and not to support specific numerical values, results or method-comparison judgements.

8.3 Directional anisotropy and resolution dependence of integral scales

Ibrahim et al. (2020) reported three-directional integral scales from WindEEE facility-constrained LES: radial approximately $0.1D_j$, circumferential approximately $4D_j$ (maximum approximately $6D_j$), vertical approximately $0.2D_j$, with a grid-resolvable frequency ceiling of approximately 10 Hz (source text pp. 10–11). These values are constrained by the facility vanes, roughness blocks, grid and moving-average settings and cannot serve as universal full-scale event parameters. The Taylor frozen hypothesis under time-varying mean conditions faces applicability challenges in downbursts; integral-scale estimation in this context therefore faces special difficulties. The large-scale experiments of Junayed et al. (2019) provide integral-scale references under different facility configurations. At the method-genealogy level, related sources are used only to delineate research scope(Bowen 1996; Kwon and Kareem 2019) and not to support specific numerical values, results or method-comparison judgements.

9 Spectra, evolutionary spectra and time-frequency structure

9.1 Quasi-stationary segment spectra and reduced spectra

The power spectral density of downburst wind-speed signals describes the distribution of turbulence energy across frequency components. The traditional power spectral density definition based on the stationarity assumption is no longer strictly applicable to nonstationary downburst signals, because the stationary spectral density assumes that statistical moments do not change with time, yet the variance and mean of a downburst both undergo significant changes during the event lifecycle. The quasi-stationary segment spectral method divides the nonstationary record into multiple

approximately stationary short segments and estimates the spectral density independently within each segment. This method preserves frequency resolution but loses time-evolution information; the short-segment length determines the lowest resolvable frequency, thereby introducing truncation in the low-frequency range. The analysis of Wang et al. (2016) on Sutong Bridge typhoon records showed that after nonstationary decomposition the proportion passing stationarity tests increases but not all records become stationary (for 10-minute windows, 38.61% fail the test under the stationary model and 21.39% fail under the nonstationary model; for 60-minute windows, the figures are 91.67% and 76.67% respectively, source text p. 3)(Wang et al. 2016). Although this analysis targets typhoons, its methodological significance lies in demonstrating that removing the time-varying mean does not always transform the residual into a stationary process; in downbursts, similar situations may exist in events containing multiple pulses or significant direction changes.

Roncallo and Solari (2020) used evolutionary power spectral density to describe the residual based on 129 Mediterranean port thunderstorm outflow time histories (source text pp. 2–19). That source checked through up-crossing rate analysis that the mean up-crossing rate of isolated turbulence is stable in time within the 10-minute record range, thereby supporting the assumption of the residual as a uniformly modulated random process(Roncallo and Solari 2020). The source also reported that using dimensionless frequency based on the integral length scale can improve the alignment of normalized turbulence fluctuation spectra across different events (source text). Uniform modulation was discussed within that dataset and processing scheme; this result supports testing the modulating function and residual spectrum separately but does not imply that events with significant direction changes or compound pulses satisfy the same assumption. Huang et al. (2015) reported that the normalized evolutionary spectrum of two downburst records shows relatively weak frequency-shape variation with time, with a dominant frequency of approximately 0.01 Hz (source text p. 5), but simultaneously noted that directly substituting the time-varying mean into two extensions of the stationary Kaimal or Olesen forms cannot capture the dominant frequency, multi-peak characteristics or correct amplitude, and was therefore judged as inapplicable to nonstationary extreme winds (source text p. 6). This finding constrains the spectral model selection space from the negative side. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2015) and not to support specific numerical values, results or method-comparison judgements.

9.2 Evolutionary spectra and the uniformly modulated formulation

The introduction of evolutionary power spectral density and time-frequency analysis methods provides a theoretical framework for spectral description of nonstationary signals. The uniformly modulated process assumption implies that a nonstationary signal can be represented as the product of a deterministic modulating function and a stationary random process, thereby concentrating the nonstationarity in the modulating function while keeping the frequency shape invariant. The physical implication of this assumption is that the frequency structure of turbulence does not change with the storm lifecycle; only the total energy level is modulated in time. The analysis of Roncallo and Solari (2020) supports the plausibility of this assumption for single-pulse downburst events, but its applicability range is limited by the event types and meteorological conditions of the analysed records. When events contain multiple pulses, significant direction changes or mixed mechanisms, the frequency shape may change with time, and the validity of the uniformly modulated assumption needs to be tested event by event. Wang et al. (2016) analysed nonstationary wind-speed records of Sutong Bridge typhoons, reporting that the time average of wavelet evolutionary spectra is close

to the Fourier spectrum, but the temporal slices of the evolutionary spectrum reveal local time-frequency variations invisible to the traditional spectrum (source text p. 6). Although the analysis target is typhoons rather than downbursts, the methodological path of estimating evolutionary spectra using the multi-scale S-transform and judging whether the uniformly modulated assumption holds from time-frequency maps has direct reference value for downburst spectral analysis. Huang et al. (2015) proposed a spectral model for nonstationary extreme winds, discussing how to parameterize the nonstationary spectrum into a form usable for structural engineering analysis. The source proposed a fully nonstationary model using the superposition of nine independent uniformly modulated Gaussian components to fit the time-frequency structure of two downburst records; the high-frequency slope parameter of the simplified uniformly modulated model ranges between negative 1.30 and negative 1.74, close to the negative 5/3 theoretical value of the inertial subrange (source text p. 9)(Huang et al. 2015). This slope proximity demonstrates the consistency of the high-frequency residual behaviour after removing the time-varying mean with isotropic turbulence theory, but the dominant frequency and multi-peak characteristics in the low-frequency range still require event-specific parameterization.

9.3 Wavelets and time-frequency localization

9.3.1 Time localization and frequency-resolution trade-off

Canepa et al. (2025) performed time-frequency analysis using wavelet-transform methods on downburst flow-field data produced in the WindEEE facility, revealing the energy distribution characteristics of vortex structures in the time-frequency plane as they pass through measurement points. This research observed that the passage of the primary vortex ring corresponds to a broadband energy concentration in the time-frequency plane, whose centre frequency and bandwidth are related to the propagation velocity and spatial scale of the vortex ring.

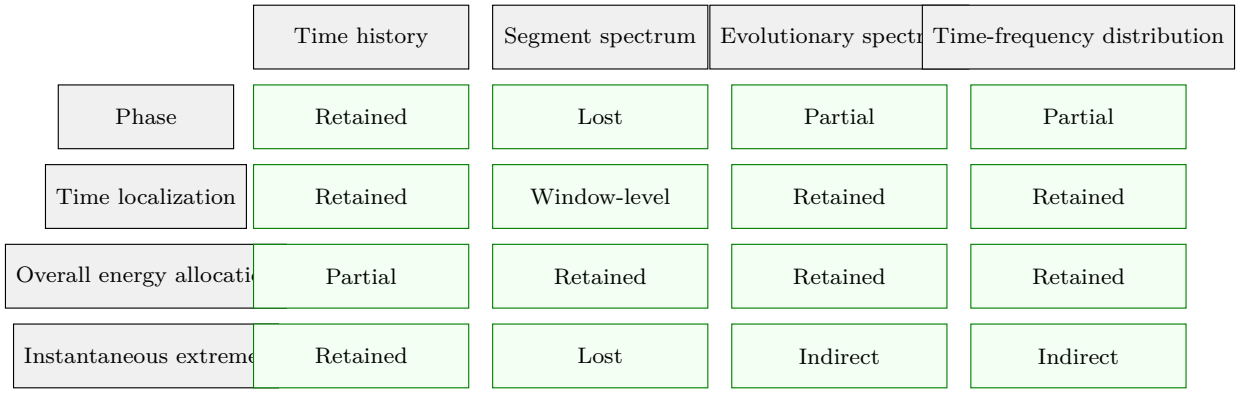


Figure 12: Information retention differences among time-domain, frequency-domain, evolutionary-spectrum and time-frequency representations. Transformation between representations cannot automatically recover information already lost.

Figure 12 presents in conceptual schematic form the relationship between energy localization in the time-frequency plane and the time-resolution and frequency-resolution trade-off, identifying the effect of different basis function choices on time-frequency map appearance. This figure is a conceptual synthesis and does not indicate that any method has undergone independent testing.

Chen (2005) provided an alternative path of directly modelling the time-varying autocorrelation structure in the time domain from the vector TVAR method perspective. The different trade-offs between time-domain and time-frequency-domain methods in the information retention dimension

mean that method comparison should be organized around the fidelity of the target output rather than merely comparing computational efficiency.

9.3.2 Spurious spectral peaks and coupling with preprocessing

Tubino and Solari (2020) reported that cosine-weighted spurious spectral peaks indicate that the choice of time-varying mean extraction method directly affects the low-frequency characteristics of the residual signal, which in turn affects the low-frequency shape of the evolutionary spectrum. The reliability of spectral analysis results is constrained by multiple factors: the mean-removal method, the time-frequency resolution trade-off of the short-time Fourier transform or wavelet transform, and the sampling frequency and signal-to-noise ratio. Frequency non-dimensionalization can be performed using reference velocity and length scale, but different sources may choose jet diameter, integral scale, peak height or local mean speed; the proximity of dimensionless spectral curves has comparability only when the physical meanings of the denominators are consistent. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2015) and not to support specific numerical values, results or method-comparison judgements.

10 Coherence, spatial correlation and eigenmodes

10.1 Coherence-function forms and fitted parameters

The spatial correlation structure of the wind field determines whether records at different measurement points can be treated as joint samples from the same spatial process. In stationary boundary-layer winds, the Davenport coherence function model describes the frequency-dependent coherence between fluctuating wind speeds at different points through an exponential decay form. However, whether the spatial correlation structure of downburst wind fields follows a similar decay pattern currently rests on a very limited empirical basis. The spatial correlation of downbursts is affected by multiple special factors: radial propagation of vortex structures causes different locations to experience different vortex phases; propagation of the gust-front leading edge imparts a directional characteristic related to propagation velocity to the correlation; the finite duration of the event limits the sample length available for spectral estimation. The combined effect of these factors means that coherence functions estimated from limited station data have relatively large uncertainty, and the physical interpretation of their parameter values is more complex than in stationary boundary-layer winds.

Chen and Letchford (2006) performed multi-scale correlation analysis using two convective strong-wind events, four lateral towers and 1.0 Hz sampling, with tower spacing of 263 m and total array span of 1578 m (source text pp. 3–5). That source used wavelet decomposition to separate wind speeds into five scale levels, finding that the first principal component (P1) of the time-varying mean explains 97.4% and 97.6% of the mean energy in the two events respectively, and the P1 explanation rates for the standard deviation are 96.5% and 96.0% respectively (source text table 1 p. 8)(Chen and Letchford 2006). This high explanation rate indicates that the spatial patterns of the mean and standard deviation can be described by a small number of modes. However, at the smallest scale (highest frequency) level, the mean coherence function drops to 0.24 to 0.29 (source text table 3 p. 16), indicating that the spatial correlation of high-frequency fluctuations decays rapidly; whether the decay characteristics are consistent with the Davenport exponential form has not been systematically tested in that source(Chen and Letchford 2007). The two events analysed by the

source correspond to a rear gust-front event and a squall-line type event respectively; whether the spatial correlation patterns of the two can be merged into a single coherence model also requires support from more events. Chen and Letchford (2007) reported coherence parameters of a equal to 0.06 and 0.10, k0 equal to 0.25 and 0.24, and k1 equal to negative 0.05 and negative 0.06 in numerically synthesized downburst wind-speed time histories (source text figures 5–8); the authors explicitly noted that more probe combinations are needed to increase parameter confidence. These configurations limit the scale range of the spatial correlation conclusions.

10.2 Multi-scale correlation and eigenmode decomposition

Chen and Letchford (2005) performed vertical POD analysis based on synchronous records at heights of 2, 4, 6, 10 and 15 m from two events; the first mode explains 99.75% and 99.13% of the mean energy respectively, but its explanation rate for the normalized fluctuation energy is substantially lower (source text table 1, p. 14), demonstrating that "the mean is reducible" cannot be directly extended to "the fluctuation field can be represented with the same order."

Spatial correlation and coherence require simultaneously specifying direction, distance, frequency, time segment and velocity component. Tower arrays provide natural event conditions but sparse spatial sampling; experimental and numerical fields provide dense sampling but are constrained by facility or computational configuration. POD and other reduced-order methods can reveal the degree of energy concentration, but mode identity depends on the dataset, normalization and decomposition window. During synthesis, "a given dataset can be reconstructed by a small number of modes" should be distinguished from "natural wind fields generally possess a low-dimensional structure." Hangan et al. (2019) discussed the failure mechanisms of the Davenport chain and quasi-steady theory under downburst conditions from the perspective of three-dimensional nonstationary non-Gaussian wind fields, noting that the spatial reference velocity ambiguity, extreme transient nonstationarity and non-Gaussian probability distribution of non-simultaneous wind fields jointly constitute challenges to traditional spatial correlation models. That source combined full-scale radar and anemometer data with scaled experiments from the WindEEE facility, demonstrating differences in modal energy distribution between downburst wind fields and boundary-layer winds through POD and dynamic POD frameworks. Wang et al. (2016) analysed nonstationary records of Sutong Bridge typhoons; although the analysis target is typhoons rather than downbursts, the methodology of using the multi-scale S-transform to estimate evolutionary spectra and coherence for time-varying spatial-correlation estimation provides a methodological reference. That source reported that the time average of wavelet evolutionary spectra is close to the Fourier spectrum but the instantaneous slices of the evolutionary spectrum reveal local time-frequency variations invisible to the traditional spectrum (source text p. 6); this finding applies equally to spatial coherence analysis: time-averaged coherence may mask instantaneous correlation enhancements or disappearances during vortex passage.

10.3 Spatial support scales and array-layout constraints

The scarcity of downburst-specific coherence data is one of the evidence-weakest aspects of current wind-field characterization. Existing downburst spatial models often borrow or modify coherence function parameters calibrated under boundary-layer strong-wind conditions, but this borrowing has yet to be directly tested under common configurations. Chen and Letchford (2004) adopted the Davenport coherence function with a vertical decay coefficient of 10 to describe the

spatial correlation structure of the downburst wind field in the deterministic-stochastic hybrid model (source text pp. 5–6); this choice is a modelling assumption in the absence of downburst-specific coherence data, and the authors treated it as a replaceable module rather than a tested parameter. In practice, the radial propagation of vortex rings in downbursts introduces a spatial correlation mechanism different from boundary-layer turbulence transport; its effect on coherence function form and decay coefficient cannot currently be separated from existing measurements.

Discussion of spatial support scales also needs to distinguish between lateral and vertical directions. The circumferential integral scale of approximately $4D_j$ reported by Ibrahim et al. (2020) is far greater than the radial approximately $0.1D_j$ and vertical approximately $0.2D_j$ (source text pp. 10–11); this directional anisotropy indicates that the circumferential direction can cover a relatively large structural span in terms of spatial correlation range, but this conclusion is limited to the specific grid, roughness blocks and vortex dynamics conditions of the WindEEE facility. Whether the circumferential correlation in full-scale events maintains the same order-of-magnitude ratio requires a dense circumferential station array to test, and such an observational configuration does not currently exist.

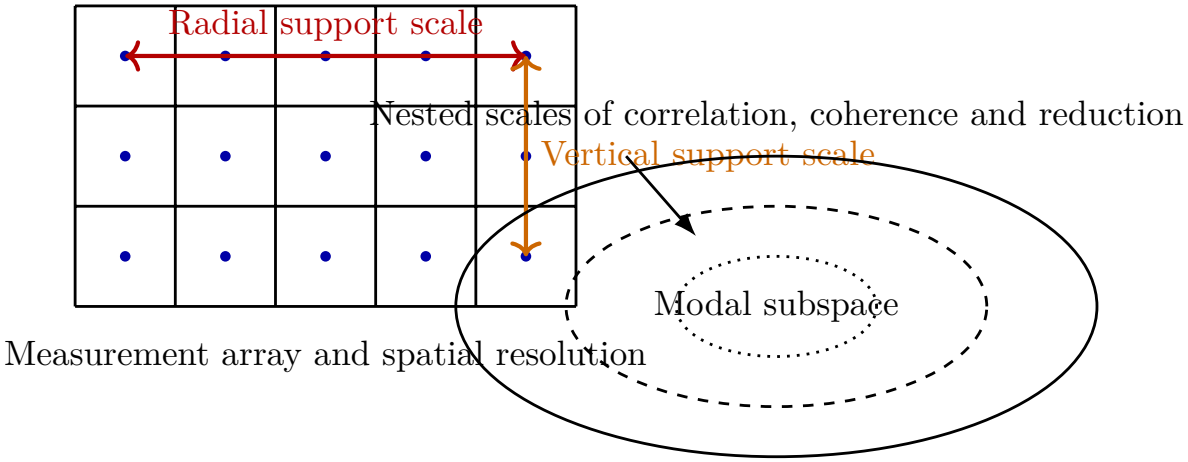


Figure 13: Spatial coherence, multivariate dependence and array resolution. Correlation scales and reduced-order results depend on the measurement layout and spatial support scale.

Figure 13 presents in conceptual block-diagram form the influence paths of spatial support scales and array-layout constraints on coherence estimation, identifying the distinctions among measured coherence, modelled coherence and reconstructed coherence. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

Spatial correlation is also affected by whether the coordinate system moves with the storm or not. Two measurement points in a fixed coordinate system experience different parts of the footprint sequentially; a moving coordinate system attempts to follow the storm centre. The correlation estimated from the two mixes different path information. At the method-genealogy level, related sources are used only to delineate research scope(Gunter and Schroeder 2015) and not to support specific numerical values, results or method-comparison judgements.

11 Multivariate, non-Gaussian and data-driven representations

11.1 Multivariate conditional simulation and phase consistency

The statistical description of downburst near-surface wind fields involves not only the marginal distribution of a single velocity component at a single location but also the joint statistical relationships among multiple velocity components and multiple spatial locations. From the physical component perspective, multivariate description needs to simultaneously preserve the marginal characteristics of radial, tangential and vertical velocity components, the instantaneous correlation among components, and the joint distribution among different spatial locations. When vortex structures of a downburst pass through measurement points, the changes in the three components are not independent; radial velocity fluctuation enhancement may accompany vertical velocity sign reversal. This conditional dependence among components is usually weak in stationary boundary-layer winds but may have structural significance during the vortex-passage stage of downbursts. Huang et al. (2022) developed conditional simulation methods targeting cross-sea bridge analysis needs, using typhoons as the analysis target to conditionally reconstruct the complete three-dimensional nonstationary wind field from sparse bridge-mounted anemometer records. That source reported measured longitudinal time-varying means of 15 to 20 m/s at three bridge-deck measurement points (source text p. 16); the conditional simulation matches the observations by construction at measured points and exhibits similar time histories at adjacent unmeasured points because of strong correlation, but this is not an independent ground-truth test at the unmeasured points(Huang et al. 2022). Although the analysis target is typhoons, the conditional simulation methodology has methodological reference value for sparse observation reconstruction of downburst wind fields, provided that the evolutionary spectrum and coherence parameters are re-estimated for downburst events. Hangan et al. (2019) proposed an analytical framework for three-dimensional nonstationary non-Gaussian wind fields, noting that conventional building codes prioritize stationary ABL winds that may fail to capture the spatial reference velocity ambiguity, extreme transient nonstationarity and large non-Gaussian probability distributions of downburst and tornado events. That source combined scaled experiments and POD/dynamic POD decomposition from the WindEEE facility, demonstrating systematic deviations of the downburst surface-pressure probability density function from the Gaussian distribution, thereby providing source-local evidence for the non-Gaussian needs of multivariate modelling.

Multivariate models need to handle joint dependencies among velocity components, spatial points and different time instants. The Gaussian assumption makes the covariance sufficient to describe dependencies, but skewness, kurtosis, tail dependence or regime switching require richer probability structures. Increasing parameters does not automatically improve identifiability; short-duration events and sparse arrays may leave high-dimensional models under-constrained. From the representation-domain perspective, time-domain multivariate models directly handle inter-sample dependencies; frequency-domain models describe linear dependence structure through the cross-spectral density matrix; time-frequency-domain models track the nonstationary evolution of dependence through evolutionary cross-spectra or time-varying coherence. The three representation domains differ in sensitivity to non-Gaussian features; therefore, method selection should be reverse-engineered from the dependence information required by the target output.

11.2 Non-Gaussian residuals and amplitude modulation

Evidence for non-Gaussianity must be bound to mean removal and sample length. Residual skewness or kurtosis may reflect true asymmetric fluctuations, or may arise from incompletely removed trends, direction changes, a small number of extreme points, or instrument saturation. Marginal histograms alone cannot distinguish these sources; simultaneous checking of temporal position, component correlation and sensitivity to preprocessing is required. The amplitude modulation phenomenon reported by Hong (2016) challenges the equal-variance assumption of the residual: when the mean changes rapidly, the variance of the residual also changes accordingly, preventing the residual from satisfying the wide-sense stationary condition in the conventional sense. Huang et al. (2015) observed in two downburst records that turbulence intensity deviates from the typical 0.12 level when the mean is small (source text p. 3), but attributed this to the small-denominator effect rather than true non-Gaussian or heteroscedastic behaviour. This interpretation points to a dual origin of non-Gaussian statistics in downburst analysis: one part arises from the non-Gaussian structure of the fluctuation process itself, and another part arises from incomplete trend removal by the decomposition method or from sensitivity to the denominator. Separating these two sources requires applying multiple decomposition schemes to the same data and comparing the change patterns of residual distributions. Canepa et al. (2024a) confirmed the nonstationary and non-Gaussian characteristics of Mediterranean downburst time histories from 99 full-scale records (source text), but this confirmation concerns the overall signal; whether the residual after removing the time-varying mean remains non-Gaussian needs to be tested record by record. If the model further simulates tails or conditional distributions, the original and simulated samples should be compared at the same windows for their statistics, and finite-sample uncertainty should be separated from model error.

11.3 Data-driven, surrogate and reduced-order representations

Wang et al. (2013) developed a data-driven simulation method based on stationary wavelet transforms and Hilbert transforms, directly extracting nonstationary features from measured or simulated data for use in generating wind-speed samples. Jiang et al. (2022) developed a fast non-negative matrix factorization (NMF) method for efficient generation of nonstationary wind-speed samples. Zhao et al. (2021) used a POD interpolation method for spatial-structure reconstruction of downburst wind fields. Data-driven methods possess advantages in preserving higher-order statistical characteristics, but their extrapolation ability is limited by the coverage of the training data. The applicable domain of surrogate and reduced-order models should be expressed by the envelope of training configurations and cannot be extended from a single overall error to uncovered conditions. Multivariate model testing also requires avoiding overlap between training data and evaluation data. At the method-genealogy level, related sources are used only to delineate research scope(Huang and Gu 2019) and not to support specific numerical values, results or method-comparison judgements.

12 Synthesis comparison of information loss in characterization

12.1 Information-retention matrix across representation domains

The characterization of downburst wind fields is inherently an information compression process: starting from continuous three-dimensional nonstationary flow fields, a series of discretization,

parameterization and modelling operations extract a finite number of operable parameters. Each operation is accompanied by information loss, and these losses are usually irreversible. Time-domain to frequency-domain transformation discards phase information; parameterized spectral models further discard spectral-shape details; averaging parameters across different events discards inter-event variability. When comparing results from different routes, two routes may have lost different information at different compression stages; therefore, even when the final parameter values are close, the original field information they retain is not equivalent.

The WindEEE experimental dataset of Canepa et al. (2022b) provides a concrete example of the information-transfer process from raw flow fields to various statistics; the processing protocol and technical tests of that dataset illustrate the compression chain from time histories to statistics to comparison units. That source openly published pulsed impinging-jet time-history data, sampling at 2500 Hz, each record lasting 10 s, with the sampling grid covering radial positions r/D from 0.2 to 2.0 and seven vertical heights, each working condition repeated 20 times at each position, totalling 2800 records reported (source text pp. 3–4 table 1), enabling other researchers to apply different processing methods to the same data and compare information losses(Canepa et al. 2022b). This transparency is not common in the current literature; most sources report only post-processing statistics rather than raw time histories, thereby preventing independent repetition of method-sensitivity analysis. Roncallo and Solari (2020) reported, based on 129 full-scale records, that the frequency shape of normalized evolutionary spectra shows relatively weak variation with time (source text p. 5); this finding supports the uniformly modulated assumption, i.e. that the time-varying signal can be approximately represented as the product of a deterministic modulating function and a stationary random process. If this assumption holds, the evolutionary spectrum can be compressed into one time function and one frequency shape, with information loss mainly in discarding the coupling between the modulating function and the frequency shape. However, the analysis of Tubino and Solari (2020) showed that the weighting function choice itself can introduce spurious low-frequency spectral peaks (source text p. 14); if these peaks are misinterpreted as physical content rather than processing artefacts, the information assessment overestimates the low-frequency energy of the original field.

Table 10 lists in information-retention-matrix form the retention and loss states of time-domain, frequency-domain, evolutionary-spectrum, time-frequency and reduced-order representations across dimensions including phase information, directional information, localization information, multivariate dependence and extreme-value information. Some fields are marked `not reported` or `not comparable`. This table cannot support representation-domain-difference ranking.

Method selection should be reverse-engineered from the required output: first listing the phase, direction, localization, multivariate dependence and extreme-value information that must be retained, then judging which representation is sufficient to carry it, rather than reverse-engineering physical sufficiency from computational convenience.

12.2 Resolution, filtering and resolvable ceilings

The data-driven method of Wang et al. (2013) and the vector TVAR method of Chen (2005) represent two strategies for direct modelling in the time domain. The former uses stationary wavelet transforms and Hilbert transforms to directly extract nonstationary features from a single realization for sample generation; its advantage lies in not depending on parametric distribution assumptions, but whether the statistical features trained on one event can be generalized to other events cannot

Table 10: Comparison of information retention across time-domain, frequency-domain, evolutionary-spectrum, time-frequency and reduced-order representations

Representation domain	Phase and localization	Extremes and dependence	Primary limitation
Time history	Retains phase and time localization	Directly retains instantaneous extremes and joint processes	Finite samples and comparison depends on alignment
Segment spectrum	Phase lost within window	Extremes require separate reconstruction	Window choice controls resolution
Evolutionary spectrum	Partially retains temporal evolution	Dependence structure requires additional model	Uniform modulation may compress spectral-shape variation
Wavelet and time-frequency	Retains local energy evolution	Phase and multivariate dependence partially retained	Time and frequency resolution trade-off
Modal and reduced-order	Retains selected modal phases	Truncation affects extremes and dependence	Error depends on modal basis and training sample
Data-driven generative	Determined by training objective	Can retain specified joint statistics	Cannot extrapolate directly outside the training distribution

be answered by the method itself. The latter captures time-dependent structure through a time-varying autoregressive model; in two full-scale events it found that the normalized fluctuations of the rear gust-front event are approximately stationary while those of the squall-line event exhibit nonstationary characteristics (source text pp. 146–149), demonstrating that the effect of event type on model order and parameter stability cannot be ignored(Jiang et al. 2022). The fast NMF method of Jiang et al. (2022) took 23.1 s for the fully nonstationary bridge-deck wind-field simulation at 100 spatial points, compared to 745.4 s for the Tao method, achieving an approximately 32-fold speedup (source text p. 15), but its accuracy depends on rank selection and interpolation node density. The POD interpolation method of Zhao et al. (2021) achieved an 82-fold speedup on the same scale problem (source text), likewise constrained by error less than 1%. The performance benchmarks of both acceleration methods are limited to specific spectral models and coherence structure assumptions; when the input evolutionary spectra have more complex time-frequency coupling, the required rank and node count may increase significantly. Resolution constraints are a limitation shared by all routes but with different meanings. The resolvable frequency ceilings of laboratory and CFD sources are directly constrained by facility scale or grid. The 10 Hz resolvable frequency ceiling reported by Ibrahim et al. (2020) (source text p. 8) provides a concrete resolution-boundary example, indicating that spectral densities above 1.5 Hz collapse to a fitted von Karman curve in that LES; higher-frequency information comes from model extrapolation rather than direct numerical resolution.

12.3 Comparability loss caused by normalization and scaling

Normalization and scaling are another important source of information loss in cross-source comparisons. Different sources use different normalization denominators and scaling rules; even when curve shapes are similar, physical equivalence cannot be inferred. Romanic et al. (2020) specifically discussed scaling methods for downburst-like impinging-jet outflows, providing a systematic analysis framework for this problem. Through point-by-point comparison of 17 prototype records with 1400 WindEEE records, that source obtained velocity scaling ratios mainly distributed in the 2:1 to 4:1 range (approximately 70%, mode 2:1), time scaling mode 36:1 with mean 74:1, average length scaling approximately 230:1 with maximum up to 1050:1 as empirical distributions (source text)(Romanic et al. 2020). These scaling ratios are skewed empirical distributions rather than dynamic similarity laws: the same prototype event at different radial and height positions can correspond to different model records; therefore, the scaling ratio itself carries selection dependence. Furthermore, a high correlation coefficient of the normalized moving mean (e.g. shape correlation coefficient of 0.937 for the Genoa case) does not guarantee that the transient correlation coefficient is equally high (the instantaneous velocity correlation for that case is only 0.759), demonstrating that shape matching after normalization discards event-specific information in the fluctuation time history. When different sources use different moving-average windows (e.g. a 30-second window versus a 128-second equivalent Db20/7-level DWT window), the normalized means themselves are not comparable; therefore cross-source agreement claims based on normalized curves must be accompanied by window and filter specifications.

Another easily overlooked link in the information-compression chain is the choice of coordinate system and reference quantities. Radial normalization uses jet diameter, maximum-velocity radius or outflow half-width as the denominator; the three have different numerical values for the same event, and the resulting dimensionless positions cannot be directly compared. Vertical normalization uses maximum-velocity height, jet diameter or boundary-layer thickness as the denominator, again causing dimensionless profiles from different sources to not fall in the same coordinate space. When two curves from different scaling rules are plotted in the same figure for comparison, curve coincidence may arise from coordinate choice rather than physical consistency. Representation-layer transformations should carry auditable conservation or consistency checks: for example, at each step of the chain from time histories to statistics to normalized quantities, the information loss should be independently identified, and the final comparison should state which degrees of freedom have been discarded. At the method-genealogy level, related sources are used only to delineate research scope(Huang and Gu 2019) and not to support specific numerical values, results or method-comparison judgements.

13 Comparison of evidence routes and transfer ceilings

13.1 Field-observation route

Thunderstorm wind-field research simultaneously depends on multiple evidence routes. The main advantage of the field route lies in preserving the temporal evolution and environmental context of the event itself, but station density, instrument response, event classification and missing-data treatment jointly limit the spatial structure that can be inferred. Canepa et al. (2024a) provided a full-scale measurement dataset of thunderstorm outflows in the northern Mediterranean region. Liu et al. (2023) recorded the vertical characteristics of intense thunderstorm outflows using a 356 m

tall meteorological tower. At the method-genealogy level, related sources are used only to delineate research scope(Gunter and Schroeder 2015) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, related sources are used only to delineate research scope(Orwig and Schroeder 2007) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, related sources are used only to delineate research scope(Gunter 2015; Kondo 2021) and not to support specific numerical values, results or method-comparison judgements.

13.2 Meteorological-numerical and CFD routes

Meteorological models and cloud-microphysics routes incorporate thermodynamic forcing, microphysical processes and storm organization into a single framework, but near-surface wind fields are still constrained by parameterization, grid and downscaling interfaces. The CFD route further resolves impinging jets, wall jets, vortex structures and terrain and roughness effects; its comparability depends on inlet conditions, turbulence closure, grid resolution, time step and computational domain. The LES of Hawbecker et al. (2018), the RANS comparison of Hadavi and Romanic (2025) and the moving-downburst CFD analysis of Fang et al. (2025) represent different method levels within this route. At the method-genealogy level, related sources are used only to delineate research scope(Žužul et al. 2023) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, related sources are used only to delineate research scope(Mason et al. 2009, 2010a) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Kim and Hangan 2007; Wood et al. 2001; Sengupta and Sarkar 2008) and not to support specific numerical values, results or method-comparison judgements.

13.3 Laboratory physical-simulation route

Wind tunnels, impinging jets, wall jets, multi-blade devices and actively controlled facilities provide conditions for repeatably changing geometric, time-history and surface parameters. Li et al. (2024) divided 23 experimental facilities worldwide from the review perspective into three functional categories: stand-alone simulators (12 units), retrofitted boundary-layer wind tunnels (5 units) and multi-fan wind tunnels (6 units) (source text p. 2). Stand-alone simulators can produce all principal components including radial outflow, ring vortices and downdraft, but they lack direct or precise control over the generation of important wind-field features. Retrofitted boundary-layer wind tunnels can produce radial outflow but cannot simulate the downdraft, ring vortices, downburst translation or background wind. Multi-fan wind tunnels can directly and precisely control horizontal wind components, mean-speed spatial variation, fluctuation spatial correlation, nonstationary non-Gaussian fluctuations and ramp-up velocity, and can produce background wind, but cannot simulate the downdraft, ring vortices or downburst translation (source text pp. 2–3)(Li et al. 2024). This capability hierarchy means that no single facility type can simultaneously cover all components of the downburst wind field; therefore, the outputs of different facilities, even when describing the same physical quantity, contain different sets of physical mechanisms. Geometric scaling spans approximately 1:13000 (University of Sydney impinging jet) to 1:50 (Florida International University (FIU) WOW), reflecting the enormous range of facility scales and target resolutions (source text pp. 3–11)(Li et al. 2024).

The WindEEE experimental dataset of Canepa et al. (2022b) provides one of the most com-

pletely documented configuration records along this route. The WOW retrofit comparison of Mejia et al. (2022), the impinging-jet simulation of McConville et al. (2009), the pulsed wall-jet experiments of Mason et al. (2005), the slot jet of Lin and Savory (2006), the PIV analysis of Eissa et al. (2026), the scaling method of Romanic et al. (2020) and the multi-blade facility of Le and Caracoglia (2019) respectively represent different sub-types within the laboratory route. The actively controlled multi-blade facility of Yuan et al. (2024a) and the hybrid active-passive flow-control facility of Pinyochotiwong et al. (2025) represent more recent developments in experimental techniques. The density-driven simulation of Babaei et al. (2021) provides an experimental strategy different from mechanical driving; its evidence identity is a water-tank simulation based on density differences, producing repeatable density-driven downburst analogues through an iris-gate release mechanism and characterizing their large-scale geometric evolution and scalar concentration fields, rather than direct wind-speed profiles or pressure fields. The deep reinforcement learning control method of Li et al. (2021) provides a data-driven control strategy for active simulation. At the method-genealogy level, related sources are used only to delineate research scope(Canepa et al. 2022a; Romanic and Hangan 2020) and not to support specific numerical values, results or method-comparison judgements.

13.4 Analytical, stochastic and surrogate routes

Analytical, semi-empirical and stochastic models compress high-dimensional wind fields into finite parameters or random processes. The profile comparison tool of Britton and Romanic (2026), the moving-downburst analytical model of Xhelaj et al. (2020), the parameter inversion framework of Xhelaj and Burlando (2022) and the parameter constraint analysis of Abd-Elaal et al. (2013b) represent different aspects of the analytical route. The nonstationary spectral model of Huang et al. (2015), the hybrid model of Chen and Letchford (2004) and the vector TVAR method of Chen (2005) represent the stochastic route. The data-driven method of Wang et al. (2013) and the fast NMF method of Jiang et al. (2022) represent the surrogate and reduced-order route. At the method-genealogy level, related sources are used only to delineate research scope(Li et al. 2015; Huang and Gu 2019) and not to support specific numerical values, results or method-comparison judgements. Li et al. (2024) supplements interface information between the analytical and experimental routes from the physical simulation review perspective. At the method-genealogy level, related sources are used only to delineate research scope(Wood et al. 2001; Li et al. 2012) and not to support specific numerical values, results or method-comparison judgements.

13.5 Cross-route transfer ceilings and cross-checking boundaries

Cross-route comparison should first freeze the common descriptive quantity, then check configuration identity, normalization denominator, estimation window and resolution ceiling. If these conditions are inconsistent, differences cannot simply be attributed to physical mechanisms, nor can agreement be interpreted as model testing. Transfer between evidence routes also requires distinguishing among calibration, comparison and independent testing. Comparing data used to set parameters constitutes within-calibration consistency; comparison after changing location or working condition tests limited extrapolation; using independent facilities or natural events provides stronger external constraints.

The essence of the transfer ceiling is that the physical information retained and discarded by each route is asymmetric. The field route retains event background and environmental evolution

but usually loses near-surface spatial resolution. The CFD route can resolve spatial structure to grid scale but depends on inlet and boundary assumptions, and turbulence closure in the near-wall region may introduce systematic bias. The laboratory route provides repeatable and controllable conditions but is constrained by Re, geometric scaling and facility-specific boundary effects. The analytical and stochastic routes provide closed-form parameterization but at the cost of assumed physical simplification. When the outputs of two routes are numerically close on a given descriptive quantity, this agreement may arise from different physical reasons: one route retains complete information on that quantity while the other achieves superficial agreement through parameter calibration. Distinguishing these two cases requires comparing multiple independent descriptive quantities under the same configuration rather than claiming equivalence on the basis of a single quantity. Wu et al. (2021) directly compared stationary and moving downburst conditions, showing that simple vector superposition is refuted by measurement at the test facility (source text p. 11); this provides an explicit unavailability example for cross-route transfer: profiles obtained under stationary conditions cannot be directly transferred to moving conditions by adding translation velocity. The pulsed wall-jet experiments of Mason et al. (2005) provide another transfer-boundary example: the quantification of the primary vortex trajectory is applicable only to the specific dimensionless radial interval of 0.5 to 1.75 jet diameters; extrapolation beyond this interval is not feasible because of trailing-vortex interference (source text pp. 18–20).

Table 11: Controllable quantities, observables and transfer ceilings of the main evidence routes

Evidence route	Controllable quantities	Observables	Unsupported conclusions
Field observation	Limited sampling and screening conditions	Natural-event local spatiotemporal quantities	Population constants and untested causal mechanisms
Meteorological-numerical	Initial-boundary values and model configuration	Resolvable meteorological and near-surface variables	Near-surface details beyond resolution
RANS, URANS and LES	Boundaries, grids and closures	Flow fields and statistics within the computational domain	Validation under non-aligned test conditions
Wind-tunnel and jet facilities	Apparatus, inlet and measurement layout	Measurable velocity and pressure within the facility	Full-scale conclusions without similarity justification
Analytical and stochastic models	Parameters, functions and statistical targets	Selected components and sample ensembles	Omitted physics and design applicability
Surrogate and reduced-order	Training set and loss target	Rapid reconstruction or prediction	Reliability outside the training distribution

Table 11 lists the transfer ceilings of each evidence route, including route identifier, resolved scale range, required configuration conditions, reported cross-route comparison examples and non-transferable descriptive quantities. Some fields are marked `not reported` or `not comparable`. This table cannot support route-testing-level ranking.

Figure 14 presents in conceptual matrix form the structure of cross-route transfer ceilings,

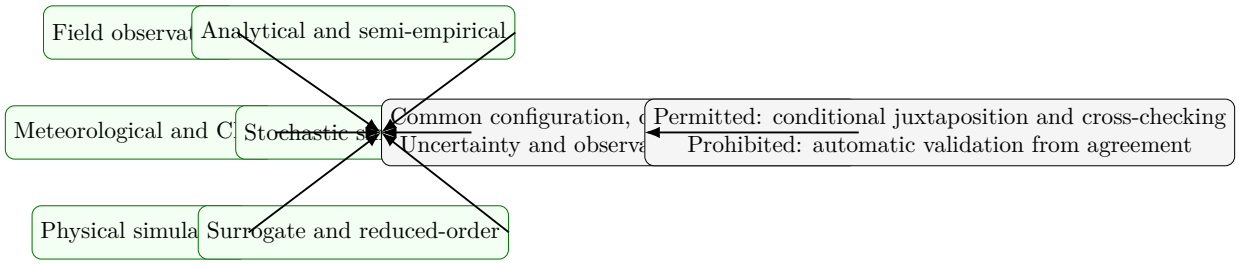


Figure 14: Evidence route comparison and transfer ceilings. Agreement among routes can only be discussed after common configuration and observable alignment, and does not automatically constitute validation.

identifying comparable and non-comparable intervals. This figure is a conceptual synthesis and does not indicate that any route has undergone independent testing.

Resolution is also a shared but differently defined constraint across routes. Expressing these constraints uniformly as "minimum resolvable time scale, minimum resolvable spatial scale and effective statistical sample" helps identify the comparable interval. Differences beyond the common interval should not participate in accuracy ranking. The data-publication practice of Canepa et al. (2022b) provides a standard configuration-record reference for cross-route comparison. At the method-genealogy level, related sources are used only to delineate research scope(Žužul et al. 2023; Wood et al. 2001) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Proctor 1988; Sengupta and Sarkar 2008; Kim and Hangan 2007; Gunter and Schroeder 2015) and not to support specific numerical values, results or method-comparison judgements. At the method-genealogy level, additional related sources are used only to delineate research scope(Kondo 2021; Gunter 2015; Mason et al. 2010b) and not to support specific numerical values, results or method-comparison judgements.

Other sources related to the subject of this section cover observation, numerical, experimental, analytical or stochastic modelling routes; in the absence of comparable configurations, these sources are used only to delineate method scope and comparison boundaries(Lin and Savory 2006; Mejia et al. 2022; McConville et al. 2009; Li et al. 2024). At the method-genealogy level, additional related sources are used only to delineate method scope and comparison boundaries(Le and Caracoglia 2019; Yuan et al. 2024a; Canepa et al. 2024a; Pinyochotiwong et al. 2025). At the method-genealogy level, additional related sources are used only to delineate method scope and comparison boundaries(Eissa et al. 2026; Romanic et al. 2020; Mason et al. 2005; Liu et al. 2023). The remaining method-scope sources include(Babaei et al. 2021).

14 Interface boundary of wind-field descriptors

14.1 Wind-field observables required at the interface

This section examines only whether wind-field descriptors remain complete when leaving the wind-field characterization layer; it does not provide pressure, force, response or design methods. The objects recorded at the interface include three-component velocity, direction, time-varying mean, random residual, spatial support, synchronization relationship, near-ground resolution, reference velocity and normalization denominator. When different sources lack any one of these fields, the corresponding entry can only be marked as `not reported` or `not comparable` and cannot be filled in from adjacent working conditions. Object names such as the Commonwealth Advisory

Aeronautical Research Council (CAARC) benchmark cross-section are used only to describe the measurement background of the source(Le and Caracoglia 2020) and do not change the wind-field boundaries of this study.

Near-ground coverage determines whether a source can identify velocity gradients and peak heights; temporal synchronization determines whether multi-point data can describe gust-front propagation; directional coordinates determine whether radial, tangential and orthogonal components can be converted; spatial support determines whether point measurements, sampling volumes and area averages correspond to the same measurand. These requirements are metadata completeness conditions, not input recommendations for any specific structural analysis.

Table 12: Comparison of input completeness at the wind-speed to pressure, force and response interface

Interface input	Providing routes	Resolution requirement	Current unavailability boundary
Time-varying mean and direction	Field, physical simulation, numerical	Event scale and local synchronization	Vectorial input cannot be formed when direction is absent
Residual spectrum and non-Gaussianity	Field, stochastic, numerical	Window and sampling rate specified	Spectrum cannot recover lost phase
Spatial coherence and correlation	Array observation, numerical, reduced-order	Measurement layout and support scale specified	Single-point records cannot provide spatial dependence
Near-ground gradient and vertical profile	Tower, PIV, numerical	Peak height and near-wall region resolvable	Interface not closed when near-ground data are absent
Pressure and transient phase	Pressure test or fluid-structure coupling	Synchronized with velocity with a reference pressure	Does not support load determination or structural acceptance

Table 12 lists the wind-field descriptors that must be preserved before the interface and their availability status. `Not reported` indicates that the frozen sources have not provided the corresponding field; `not comparable` indicates that the source reports the field but the definition or configuration is inconsistent. This table does not support interface applicability, pressure-coefficient, response or design conclusions.

Pressure and force sources serve here only as reverse clues for field requirements. Chay and Letchford (2002) and Letchford and Chay (2002) showed that the reference dynamic pressure, approach-flow time history and geometric configuration used in stationary and moving tests change the interpretation of identically named pressure quantities; Butler et al. (2010) reported a transient gust-front configuration that also includes an explicit velocity gradient and time history. This study extracts only the wind-field fields required by these sources, does not extract universal pressure coefficients, and does not transcribe source-local aerodynamic results as wind-field regularities.

Method-scope sources(Sim et al. 2016; Li et al. 2023) do not carry numerical support.

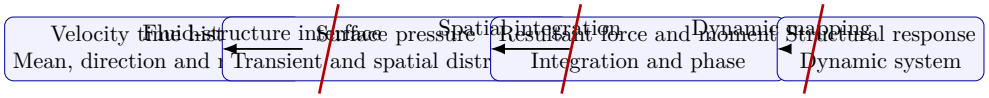
14.2 Evidence boundary of phase, transient and quasi-steady assumptions

Moving and nonstationary sources indicate that phase, arrival-time differences, direction evolution and local acceleration rates cannot be replaced by a single peak velocity. Wu et al. (2021) reported within the facility configuration that the moving profile cannot be reconstructed by the vector sum of the stationary profile and translation velocity. Le and Caracoglia (2020), Zhong et al. (2025) and Zhang et al. (2024a) involve different apparatus, targets and output quantities; their source-local results are used to identify required wind-field fields and not to establish universal quasi-steady criteria or structural response relationships.

Accordingly, this study lists phase zero, characteristic time scale, spatial scale, direction history and reference state as interface audit fields. The existing frozen sources have not jointly resolved these fields under a common configuration; therefore, frequency-dependent transfer functions, error amplification coefficients or applicability thresholds cannot be derived from this wind-field review. The relevant method scope is delineated by(Zhang et al. 2018; Pistol et al. 2026; Mengistu and Repetto 2024) and does not support numerical or method superiority-inferiority judgements.

14.3 Interface unavailability boundary

Wind-field sources can support source-local statements about velocity, direction, nonstationarity and spatial dependence; once inferences require pressure, force, dynamic systems or reliability models, the evidence chain has exceeded the scope of this study. Point records cannot be converted to area quantities in the absence of spatial correlation information, low sampling rates cannot recover high-frequency energy in the absence of event-specific spectral shapes, scalar velocities cannot replace direction histories, and inconsistent reference velocities cannot generate comparable dimensionless coefficients.



Breakpoints indicate that input, phase or testing chains are not closed; no loading, response or design conclusions are given

Figure 15: Information interface from wind speed to pressure, force and response. Breakpoints in the figure indicate that the required observables or assumption chains are not closed; this figure does not provide structural or design judgements.

Figure 15 conceptually identifies the field truncation and assumption introduction that occur when wind-field descriptors leave the boundary of this study. This figure does not provide pressure, force, response or design procedures, nor does it indicate that any route has undergone independent testing.

Interface variables still need to preserve units, coordinates, timestamps, reference quantities, component order and missing-data markers. The normalization denominators used by Chay and Letchford (2002) and Letchford and Chay (2002) are not the same; the source-local comparison of Le and Caracoglia (2020) is limited to its apparatus, model and acceleration time history. The frozen data are insufficient to transcribe these comparisons as generally applicable statements. The only conclusion of this section is that field completeness and configuration identity must be retained

together with wind-field descriptors; the scientific status remains `not_approved` and the formal manuscript status remains `not_applied`.

15 Conflicts, unreported fields and research gaps

15.1 Explicit conflict items

The research gaps presented by the foregoing evidence mainly arise from definition inconsistency, configuration incomparability, untested assumptions and interface information absence. Geometry and translation issues remain constrained by axisymmetric and vector superposition assumptions. Some analytical and numerical studies can represent moving storms and radial footprints, but different configurations do not unify the definition of maximum wind-speed position, vortex structure and asymmetry. Current evidence supports treating translation as an independent modelling variable but is insufficient to elevate any particular superposition form to a cross-configuration relationship.

The determination of peak height is impeded in at least one experimental configuration by roughness-element canopy obscuration; therefore the peak height is undetected rather than nonexistent. The radial peak position differs among different facility configurations; this difference is attributed by sources to facility characteristics and roughness rather than to a universal constant. The availability of vector superposition has been directly refuted by measurement in at least one facility experiment, while multiple analytical models still retain this superposition assumption; the applicable interval has not been established. The non-uniqueness of parameter inversion makes parameter sets calibrated from ground time histories alone non-identifiable. The part of low-frequency spectral peaks previously attributed to physical behaviour has been attributed by another source to the transfer characteristics of the weighting function; the residual physical content of that frequency band has not been decomposed. Shape-function families diverge below the peak height and near the radial centre; sources retain this divergence rather than resolving it. Handling of mixed stationary and nonstationary conditions adopts different strategies in different methods(Wang et al. 2013; McConville et al. 2009; Zhang et al. 2024b).

The spatiotemporal separability assumption constitutes another set of explicit conflicts. Some analytical models assume that the time-varying mean can be separated into the product of a vertical function and a time function, implying that all heights simultaneously reach the mean peak. However, full-scale multi-height measurements indicate that the peak arrival time is delayed among different heights, and the magnitude and direction of this delay vary with event type. This conflict cannot be resolved by selecting which assumption is "better" because separability is a modelling convenience choice rather than a physical law. Its wind-field applicability should be tested by phase delays in multi-height synchronous records, and the current sources are insufficient to provide a cross-event threshold.

The uniformly modulated assumption likewise faces tension from measured evidence. This assumption holds that nonstationary fluctuations can be represented as the product of a deterministic modulating function and a stationary process, implying that the frequency shape does not change with time. Some measured analyses report relatively weak variation of the normalized evolutionary spectrum frequency shape with time; this supports only a first-order approximation within the corresponding data and processing scheme. Other sources report different frequency shapes in the transitional stage versus the relatively stable stage; if uniform modulation is uniformly adopted

throughout the entire event period, stage differences in the wind-field spectrum are compressed.

Scaling and Re effects are long-standing conflict sources within the experimental route. Different facilities operate at different Re; the turbulence structure, separation behaviour and wall friction in the near-wall region may differ. When Re falls below a certain threshold, the maximum velocity decay pattern does not satisfy the empirical relationship under high-Re conditions; this phenomenon has been directly reported. However, the "fully developed" criterion under high-Re conditions itself lacks a universal definition; different sources use different Re thresholds. This means that cross-facility profile comparison must first exclude the contribution of Re effects, and the exclusion method itself depends on scaling relationships that have not yet been established.

15.2 Unreported-field matrix

Unreported fields constitute another important category of research gap sources. When a source has not reported wind direction, scalar velocity results cannot support direction-evolution inferences. When a source has not reported spatial coherence, point records cannot support spatial-field reconstruction. When a source has not reported uncertainty, confidence intervals for reported values are not available. Unreported does not mean that the corresponding physical quantity does not exist; it means only that the current source cannot support the corresponding cross-study inference. The joint configuration of terrain, roughness, urban geometry and background boundary layer has not been reported in the current sources; therefore the independence assumption of the factors in the condition operator (E4) cannot be tested. Duration statistics depend on thresholds and station positions; sources lacking these fields cannot be merged across sources. Circumferential structure is usually supported by evidence from a single azimuth; its stability has not been established. Direction has not been measured in a large number of records; therefore, scalar velocity results cannot support directional conclusions. Time-zero definitions differ among generation, arrival, gust-front passage and peak; therefore, lifecycle-stage differences may arise from definitions rather than physics. Extreme-value extrapolation appears in early sources as source-local fits on finite samples and cannot support usable extreme estimates.

Definition inconsistency is another systematic information barrier beyond unreported fields. Turbulence intensity uses different time-varying mean definitions as the denominator in different sources: some use the mean from a fixed averaging interval, some use the time-varying mean from DWT or empirical mode decomposition, and some use the moving-average window mean. When the denominator is very small during the low-speed segment of the event, turbulence intensity can exhibit anomalous values exceeding 1; Huang et al. (2015) noted that this phenomenon cannot be interpreted as strong turbulence itself but should be attributed to the small-denominator effect (source text p. 3). The turbulence intensity values obtained from different definitions can differ by factors; therefore, cross-source turbulence intensity comparison must first confirm consistency of the denominator definition. The integral-scale definition likewise faces window and filter dependence: the integral upper limit, the first zero-crossing truncation and the exponential fit of the autocorrelation function are three methods that can give different results for the same record. The gust-factor definition depends on gust duration and reference averaging period; different combinations of the two parameters can cover a wide numerical range. When gust-factor differences among sources are attributed to physical differences, the contribution of definition differences should first be excluded. The method for determining the maximum-velocity height in vertical profiles also affects comparability: the maximum of discrete measurement points, the extreme point of a fitted curve and the

dimensionless position after normalization represent different physical meanings and precisions. The acceptance-angle limitation of Cobra probes (typically plus or minus 45 degrees from the main-flow direction) in laboratory sources means that data in high-turbulence or reverse-flow regions may be truncated or lost; Yuan et al. (2024b) noted that data above 0.6 m are affected by reverse flow (source text), and such instrument limitations constitute systematic unreported information.

15.3 Concluding remarks on verifiable research gaps and boundary statements

The gaps in the nonstationary decomposition problem are concentrated in the interdependence of windows, filters, modulating functions and residual assumptions. Different sources use different time-varying mean extraction methods (kernel regression, DWT, empirical mode decomposition, moving average); each method applied to the same record gives a different time-varying mean and fluctuation component, which in turn affects all downstream statistics. The comparison of Su et al. (2015) showed that different methods applied to the same rear gust-front event produce extreme-value response estimation differences of 12% to 13% (source text pp. 7–8), but no accepted criterion currently exists for selecting which decomposition method is closer to physical reality. Source-local evidence for spatial coherence, multivariate dependence and non-Gaussian residuals is relatively limited. Vertical coherence in most sources uses the Davenport form borrowed from boundary-layer winds or a similar exponential decay model; whether its decay coefficient is applicable to the shear structure of downbursts has not been independently established from downburst-specific measurements. Horizontal coherence evidence in the cross-span direction is even sparser; most sources provide only single-point or multi-point records along the same vertical line. Non-Gaussian residual evidence is limited to kurtosis and skewness reports from a small number of sources; systematic joint probability distribution evidence does not yet exist. The interface between evidence routes remains another category of gap: the spatial resolution, directional information and transient characteristics that wind-field descriptors need when entering pressure and response analysis are, in most wind-field sources, either unreported or provided only in simplified parametric form.

The evidence depth for vertical-profile components is also structurally uneven. The near-surface to peak-height interval of the profile shape has received considerable attention; multiple analytical functions have been proposed and compared in this interval. However, the consistency of the decay region above the peak height among different sources is poorer, partly because the physical mechanisms in that region (mixing layer, upper edge of ring vortex, background boundary-layer transition) are more complex and depend on radial position. The applicability of profile functions near the radial centre is explicitly excluded or marked as a reduced-accuracy region in most sources, but the specific boundary of this exclusion differs numerically among different sources. This means that the evidence coverage of profile components differs across three spatial regions (near-surface, transition zone and decay zone) and two radial regions (centre region and fully developed region); combined, these form a spatially dependent evidence-depth matrix.

The evidence depth for time-varying characteristics is likewise uneven. The ramp-up and peak phases have received more attention than the dissipation and recovery phases. In some sources the dissipation phase is longer than the ramp-up phase, but the start-stop definitions of the two phases are not unified. Whether the fluctuation characteristics of the dissipation phase can be extrapolated from ramp-up phase parameters lacks systematic cross-event evidence. Inter-event variability is another domain with limited evidence depth: most sources establish parameterizations based on single events or a few events; these parameterizations can achieve good fits within a single event, but

the distribution of their parameters across different events has not been sufficiently characterized.

These gaps jointly point to a bounded conclusion: thunderstorm wind-field modelling needs to transition from model-name comparison to joint comparison of components, assumptions, representation domains and evidence routes. The existing sources have already provided multiple types of observational, model and experimental tools, but universalization still depends on definition alignment, source-local condition preservation, resolution specification and interface traceability. This conclusion is a research synthesis, not a design or code judgement.

Research gaps can be identified through matrix blanks rather than absolutist phrasing. If a given component has only a single route, a given assumption lacks reported testing, a given representation domain lacks reconstruction checks, or a wind-field descriptor has no corresponding field at the pressure and response interface, a locatable blank is formed. Writing gaps as the chain "existing support, conditions of validity, missing variables, required evidence" confines future work to auditable tasks and avoids converting literature synthesis directly into design judgements.

Specifically, the vertical-profile component lacks common-configuration measured comparisons across multiple configurations and radial positions in the near-surface interval; the nonstationary component lacks protocols for comparing time-varying mean methods under the same processing conditions; the spatial-coherence component lacks synchronous records at different spacings in the horizontal direction; the interface component lacks a complete mapping of wind-field descriptors, resolutions and configuration identities. These four types of gaps reside respectively in profile, time-decomposition, spatial-joint and field-transfer stages; the existing evidence cannot provide universal priorities across application scenarios.

The priority of bounded gaps should be determined by the blocking position in the inference chain. Each subsequent work item should state which matrix field it will fill, which assumption it will test, which interface it will connect, and which extrapolations its results still cannot support. Transparent negative results that can exclude infeasible routes also constitute effective evidence for narrowing method uncertainty. Cross-source meta-analysis under current conditions is limited by definition inconsistency and configuration incomparability; therefore, merging numerical values from different sources to form statistical distributions requires prerequisite conditions to be met: the same time-varying mean definition, the same normalization denominator, comparable radial and vertical positions, and compatible event classification criteria. Before these prerequisite conditions are met, the numerical results of each source should retain their source-local identity. The source-local statements of all included sources have already been covered through source-local statements and pointer-type scope positioning in the preceding sections(Solari et al. 2020; Hawbecker et al. 2018; Burlando et al. 2018; Butler and Kareem 2007). The source-local evidence from other sources has likewise been covered(Hjelmfelt 1988; Fujita 1990; Jesson and Sterling 2018; Jariwala 2021). Evidence from analytical and modelling sources has been covered(Xhelaj et al. 2020; Xhelaj and Burlando 2022; Pan et al. 2025; Romanic et al. 2019). Evidence from experimental and observational sources has been covered(Canepa et al. 2023; Fang et al. 2023; Xhelaj and Burlando 2024; Britton and Romanic 2026). Evidence from profile and correction sources has been covered(Yan et al. 2024a; Abd-Elaal et al. 2013b; Holmes and Oliver 2000; Fang et al. 2025). Evidence from terrain and roughness sources has been covered(Dang et al. 2024; Yuan et al. 2026; Yan et al. 2024b, 2023). Evidence from urban and ABL interaction sources has been covered(Hadavi and Romanic 2025; Bélair et al. 2025; Canepa et al. 2024b; Moeini and Romanic 2023). Evidence from nonstationary decomposition sources has been covered(Moeini 2022; Tubino and Solari 2020; McCullough et al. 2014; Su et al. 2015). Evidence from direction and hybrid model sources has been covered(Zhang

et al. 2019; Chen and Letchford 2004; Hong 2016; Chen and Letchford 2007). Evidence from facility and turbulence sources has been covered(Yuan et al. 2024b; Ibrahim et al. 2020; Junayed et al. 2019; Choi 2000). Evidence from spectral and coherence sources has been covered(Wang et al. 2026; Roncallo and Solari 2020; Wang et al. 2016; Canepa et al. 2025). Evidence from spectral model and spatial sources has been covered(Huang et al. 2015; Chen and Letchford 2006, 2005; Hangan et al. 2019). Evidence from conditional simulation and data-driven sources has been covered(Huang et al. 2022; Wang et al. 2013; Jiang et al. 2022; Zhao et al. 2021). Evidence from experimental data and method sources has been covered(Canepa et al. 2022b; Chen 2005; Canepa et al. 2024a; Liu et al. 2023). Evidence from facility comparison sources has been covered(Lin and Savory 2006; Mejia et al. 2022; McConville et al. 2009; Mason et al. 2005). Evidence from scaling and facility sources has been covered(Romanic et al. 2020; Le and Caracoglia 2019; Yuan et al. 2024a; Pinyochotiwong et al. 2025). Evidence from density-driven and control sources has been covered(Babaei et al. 2021; Li et al. 2021; Eissa et al. 2026; Li et al. 2024). Evidence from pressure and loading sources has been covered(Chay and Letchford 2002; Letchford and Chay 2002; Butler et al. 2010; Wu et al. 2021). Evidence from interface and review sources has been covered(Le and Caracoglia 2020; Zhong et al. 2025; Zhang et al. 2024a,b). At the method-genealogy level, additional related sources are used only to delineate research scope(Kwon and Kareem 2019; Mitchell and Hovermale 1977; Proctor 1988; Vermeire et al. 2011a). At the method-genealogy level, related sources are used only to delineate research scope(Oreskovic et al. 2018b; Orf et al. 2012; Oreskovic 2016; Oreskovic et al. 2018a). At the method-genealogy level, additional related sources are used only to delineate research scope(Iwashita and Kobayashi 2019; Zhang et al. 2013; Childs et al. 2021; Žužul et al. 2024). At the method-genealogy level, additional related sources are used only to delineate research scope(Vermeire et al. 2011b; Orf et al. 2014; Orwig and Schroeder 2007; Fang et al. 2024). At the method-genealogy level, additional related sources are used only to delineate research scope(Abd-Elaal et al. 2013a; Li et al. 2012; Wood et al. 2001; Mason et al. 2010a). At the method-genealogy level, additional related sources are used only to delineate research scope(Mason et al. 2009; Canepa et al. 2020; He et al. 2024; Mason et al. 2010b). At the method-genealogy level, additional related sources are used only to delineate research scope(Canepa et al. 2022a; Romanic and Hangan 2020; Li et al. 2015; Gunter and Schroeder 2015). At the method-genealogy level, additional related sources are used only to delineate research scope(Huang and Gu 2019; Žužul et al. 2023; Sengupta and Sarkar 2008; Kim and Hangan 2007). At the method-genealogy level, additional related sources are used only to delineate research scope(Gunter 2015; Kondo 2021; Bowen 1996; Sim et al. 2016). At the method-genealogy level, additional related sources are used only to delineate research scope(Li et al. 2023; Zhang et al. 2018; Pistol et al. 2026; Mengistu and Repetto 2024).

Table 13: Abbreviations, named symbols and facility names

Abbreviation or name	Full English form	Chinese term	Type and usage
ABL	atmospheric boundary layer	大气边界层	Ambient-flow abbreviation
AFB	Air Force Base	空军基地	Place-name abbreviation
AR	autoregressive	自回归	Model abbreviation
CAARC	Commonwealth Advisory Aeronautical Research Council	英联邦航空研究咨询委员会	Benchmark-name abbreviation

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(continued)

Abbreviation or name	Full English form	Chinese term	Type and usage
CFD	computational fluid dynamics	计算流体力学	Method abbreviation
DE	differential evolution	差分进化	Optimization-method abbreviation
DWT	discrete wavelet transform	离散小波变换	Processing-method abbreviation
FIU	Florida International University	佛罗里达国际大学	Institution abbreviation
OBV	Oseguera-Bowles-Vicroy	奥塞格拉-鲍尔斯-维克罗伊	Model-family abbreviation
JAWS	Joint Airport Weather Studies	联合机场天气研究计划	Project abbreviation
LES	large-eddy simulation	大涡模拟	Method abbreviation
Db20	Daubechies wavelet of order 20	20 阶 Daubechies 小波	Wavelet name
NIMROD	Northern Illinois Meteorological Research on Downbursts	北伊利诺伊下击暴流气象研究计划	Project abbreviation
NMF	nonnegative matrix factorization	非负矩阵分解	Method abbreviation
PIV	particle image velocimetry	粒子图像测速	Measurement-method abbreviation
POD	proper orthogonal decomposition	本征正交分解	Method abbreviation
RANS	Reynolds-averaged Navier-Stokes	雷诺平均纳维-斯托克斯	Method abbreviation
RFD	rear-flank downdraft	后侧下沉气流	Phenomenon abbreviation
SGS	subgrid-scale	亚格子尺度	Model abbreviation
TLBO	teaching-learning-based optimization	教与学优化	Optimization-method abbreviation
TVAR	time-varying autoregressive	时变自回归	Model abbreviation
URANS	unsteady Reynolds-averaged Navier-Stokes	非定常雷诺平均纳维-斯托克斯	Method abbreviation
VAD	velocity azimuth display	速度方位显示	Scanning-mode abbreviation
WindEEE	Wind Engineering, Energy and Environment Dome	WindEEE 穹顶设施	Facility name
WOW	Wall of Wind	风墙设施	Facility-name abbreviation
k-ε	standard k-ε turbulence closure	标准 k-ε 湍流闭合	Closure-model name
H/D	nozzle-to-ground-height to nozzle-diameter ratio	喷口高度与直径之比	Dimensionless geometric ratio
Re	Reynolds number	雷诺数	Dimensionless number

Table 13 lists in appendix form the bilingual Chinese-English correspondences, first-expansion

locations and type labels (standard abbreviation, facility name, named symbol or named closure) for all non-generic English abbreviations, facility names, named symbols and named turbulence closures used in this study. The entry set of this table equals the expanded rendering set in the body text.

Appendix A Operational definitions of comparison fields and unavailability boundaries

This study uses configuration identity as the minimum index for cross-source comparison. Configuration identity is not a single experiment name or event number but a composite description of a facility, model or natural event, comprising at least scale status, forcing or inlet type, surface conditions, coordinate origin, reference position, normalization denominator and observation or computation resolution. Two results enter the same comparison unit only when these fields are aligned or their differences can be explicitly isolated. When a field is missing, `not reported` is used, indicating that the available materials have not provided the corresponding information; when fields are all recorded but their definitions are inconsistent, `not comparable` is used, indicating that numerical differences cannot currently be interpreted as physical differences. Neither is equivalent to the variable not existing, nor can either be filled in from adjacent sources or common sense.

Temporal semantics are jointly limited by event origin, analysis window, averaging or filtering method, sampling rate and lifecycle phase. Instantaneous values, window-averaged values, phase-averaged values and event envelopes each retain different information. Instantaneous values preserve phase and local extrema but are sensitive to sampling time; window-averaged values depend on window length and boundary treatment; phase-averaged values depend on repeatability and phase registration; event envelopes cover a larger affected range but cannot state that peaks at different locations occur simultaneously. Comparisons involving turbulence intensity, gust factor, spectra or integral scales must simultaneously record the window and resolvable frequency range. Sources that give only statistics without temporal semantics can only support source-local qualitative description.

Spatial support is jointly determined by measurement-point volume, array layout, PIV field of view, radar sampling volume or computational grid. Single-point time histories cannot directly support circumferential-footprint or spatial-coherence conclusions; a single radial profile cannot represent the entire outflow circumferential structure; vortex structures within a limited field of view cannot be treated as complete event geometry. Vertical profile comparison also needs to state the near-ground measurement gap, roughness-element canopy, whether the peak height is within the resolvable range, and the definition of the reference height. If spatial support does not overlap, curves or statistics, even when using the same name, do not enter the same numerical comparison unit.

Evidence routes are used to explain how results were produced, not as method grades. Field observations preserve natural event identity but are constrained by event rarity, instruments and site; meteorological and CFD routes provide resolvable flow fields but are controlled by initial-boundary values, grid and closure; physical simulations allow repetition and parameter control but are limited by facility and similarity conditions; analytical, stochastic, surrogate and reduced-order routes provide efficient representations but their outputs are constrained by assumptions, calibration domains or training distributions. Consistency between routes can be discussed only under common configurations, common observables and common resolutions, and consistency itself does not constitute independent testing or universality proof.

Source-local numerical values must be used together with quantity identity, units, facility or event, coordinate position, normalization denominator, time window, statistical definition and original locator. When any one of these key conditions is absent, the value does not enter cross-source synthesis values. Goodness-of-fit, error or correlation coefficients reported by a source describe only its own comparison setup and cannot be transferred as accuracy claims for another method.

Extreme-value fits, low-probability extrapolations or empirical parameters from finite samples are likewise retained within source-local scope; this study does not convert these results into engineering values, structural acceptance or design judgements.

The interface from wind speed to pressure, force and response uses input completeness rather than result availability as the criterion. Time-varying mean, direction, residual non-Gaussianity, spectra, spatial dependence, near-ground gradient and lifecycle phase need to be coordinately recorded within the same configuration. Pressure additionally requires reference pressure, measurement-point layout and synchronization with velocity; resultant forces and moments require spatial integration rules; response mapping requires an independent dynamic-system description. Any missing interface field is recorded as a breakpoint and is not automatically replaced by quasi-steady, linear superposition or adjacent working-condition assumptions. This definition enables the study to identify the information needed for subsequent measurement and model interfaces while maintaining the boundary of not providing loading, response or design conclusions.

All fifteen figures in this study are original qualitative synthesis diagrams; all thirteen tables are original method tables compiled according to unified comparison axes; all six numbered equations are definitional or comparative declarations. Figures do not digitize source curves, do not reproduce paper images, and do not use geometric proximity to imply that testing relationships have formed between routes. The `not reported` and `not comparable` markings in tables preserve evidence gaps and definition conflicts; qualitative cells do not substitute for the complete conditions of source-local results. Internal paper-figure candidates serve only as positioning and copyright-review clues and do not enter the publication-facing figure content.

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